細菌性疾病 (I)

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細菌性疾病 (I)

* 胸膜肺炎放線桿菌
* 萎縮性鼻炎
* 副豬格拉氏菌
* 大腸桿菌感染

胸膜肺炎放線桿菌
(*Actinobacillus pleuropneumoniae*)

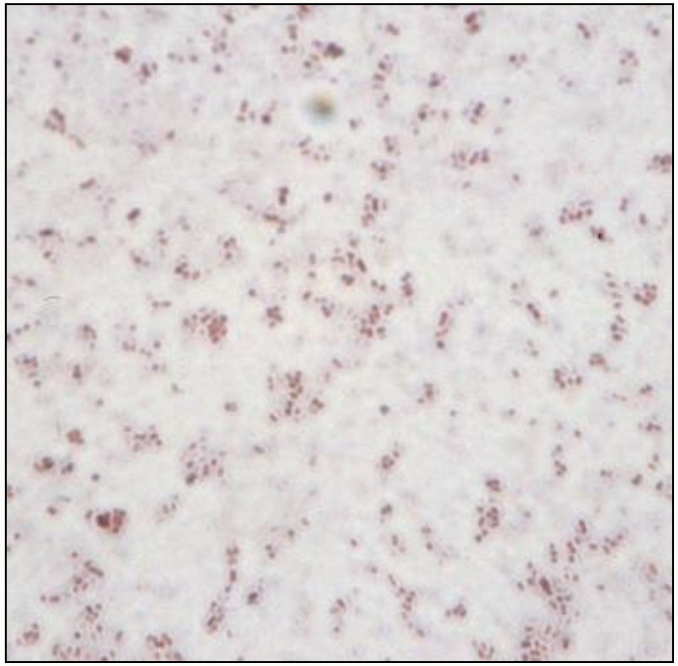
關聯 (Relevance)

- *Actinobacillus pleuropneumoniae* (*A. pleuropneumoniae*, APP) 造成豬隻胸膜肺炎(pleuropneumonia)
- APP is one of the most important bacterial respiratory pathogens of swine worldwide.
- It causes severe, rapidly progressive fibrinohemorrhagic and necrotizing pleuropneumonia (纖維出血及壞死性胸膜肺炎).

病因(Etiology)

* *Actinobacillus pleuropneumoniae*

Gram-negative, non-motile, encapsulated, facultatively anaerobic coccobacilli/short rods.



Source: microbiologyatlas.kvl.dk

* APP可依據莢膜多醣體 (capsular polysaccharide) 及脂多醣體 (lipopolysaccharide) 上的差異，區分成19種血清型 (serotype)。

* 根據生長是否需要菸鹼醯胺腺二核苷酸(nicotinamide adenine dinucleotide, NAD or V factor)可分為2種生物型別 (biotype)

    Biotype I: require NAD (17 serotypes: 1-12, 15-19)

    Biotype II: do not require NAD (8 serotypes: 2, 4, 7, 9, 11, 13, 14, 17)

Taiwan update: Among recent clinical isolates (2015–2022), serovar 15 was predominant, followed by serovars 5, 1, 7 and 2. (https://pubmed.ncbi.nlm.nih.gov/40759949/)

Actinobacillus pleuropneumoniae on chocolate agar



NAD (V factor)-dependent isolates grow well on chocolate agar.

公共衛生(Public health)

- *A. pleuropneumoniae* is considered host-adapted to pigs and is not regarded as a zoonotic pathogen.

流行病學 (Epidemiology)

- Worldwide distribution; predominant serovars vary geographically.
- Some serotypes are prevalent in certain countries and several serotypes often occur in the same country (or within farms).
- Some serotypes are considered to be of low virulence and of no epidemiologic importance in certain countries, but may be epidemic in others.

流行病學(Epidemiology)

- APP infects only the respiratory tract of pigs.

- In peracute (甚急性) and acute pulmonary infections, it can be found not only in pneumonic lung, but also in large numbers in nasal discharge. Survivors may remain carriers for several months, mainly in chronic lung lesions and tonsils.

- Transmission between herds occurs through the introduction of carrier animals to naïve population. The main route of spread is by direct nose-to–nose contact within short distance. Moving and mixing pigs increases the risk of spread.

流行病學(Epidemiology)

- In infected herds, piglets may acquire APP early in life through close contact with carrier sows.

- Survival of the organism in the environment is of short duration, especially in warm, dry conditions.

流行病學(Epidemiology)

- There has been an increase in clinical APP in many European countries in recent years as reflected in increased chronic pleuritic adhesions in abattoirs.

- This is likely due to the older weaning ages mandated by new welfare laws, since APP is known to be a late colonizer in piglets and earlier weaning dramatically reduces carriage rates in weaned pigs.

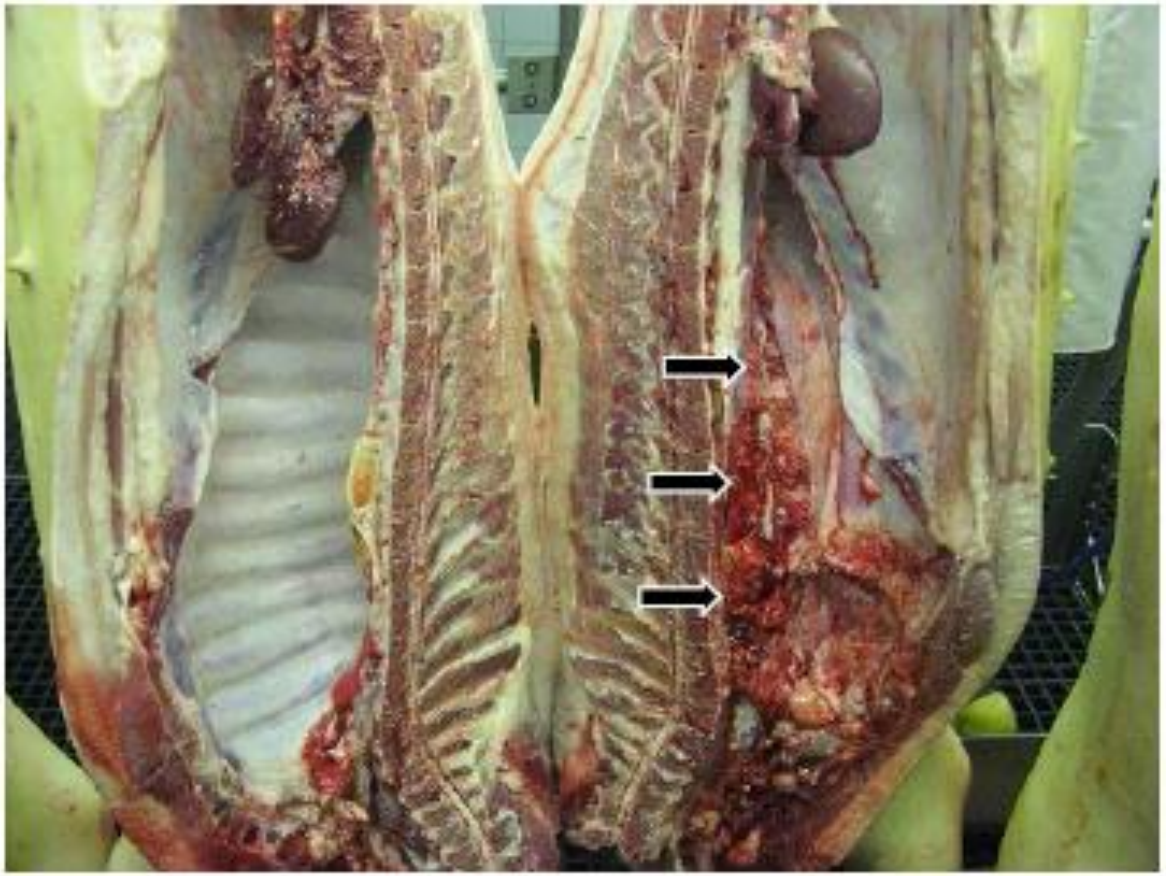


Figure 43.6 Split carcass hanging vertically. Note that the right lung remains adhered in the right thoracic cavity (arrows), whereas the left lung is removed, revealing the empty left thoracic cavity devoid of fibrinous adhesions. *Source:* Courtesy of Dr. Scott Hurd.

Factors influencing disease expression:

* Virulence of the strain
* Crowding
* Adverse climate conditions
* Rapid changes in temperature
* High relative humidity
* Insufficient ventilation
* Exacerbated by the diseases like pseudorabies (PR) or porcine reproductive and respiratory syndrome (PRRS)

致病機制 (pathogenesis)

* Four basic stage:

1. Colonization: fimbriae, LPS

2. Nutrient uptake: iron, nickel

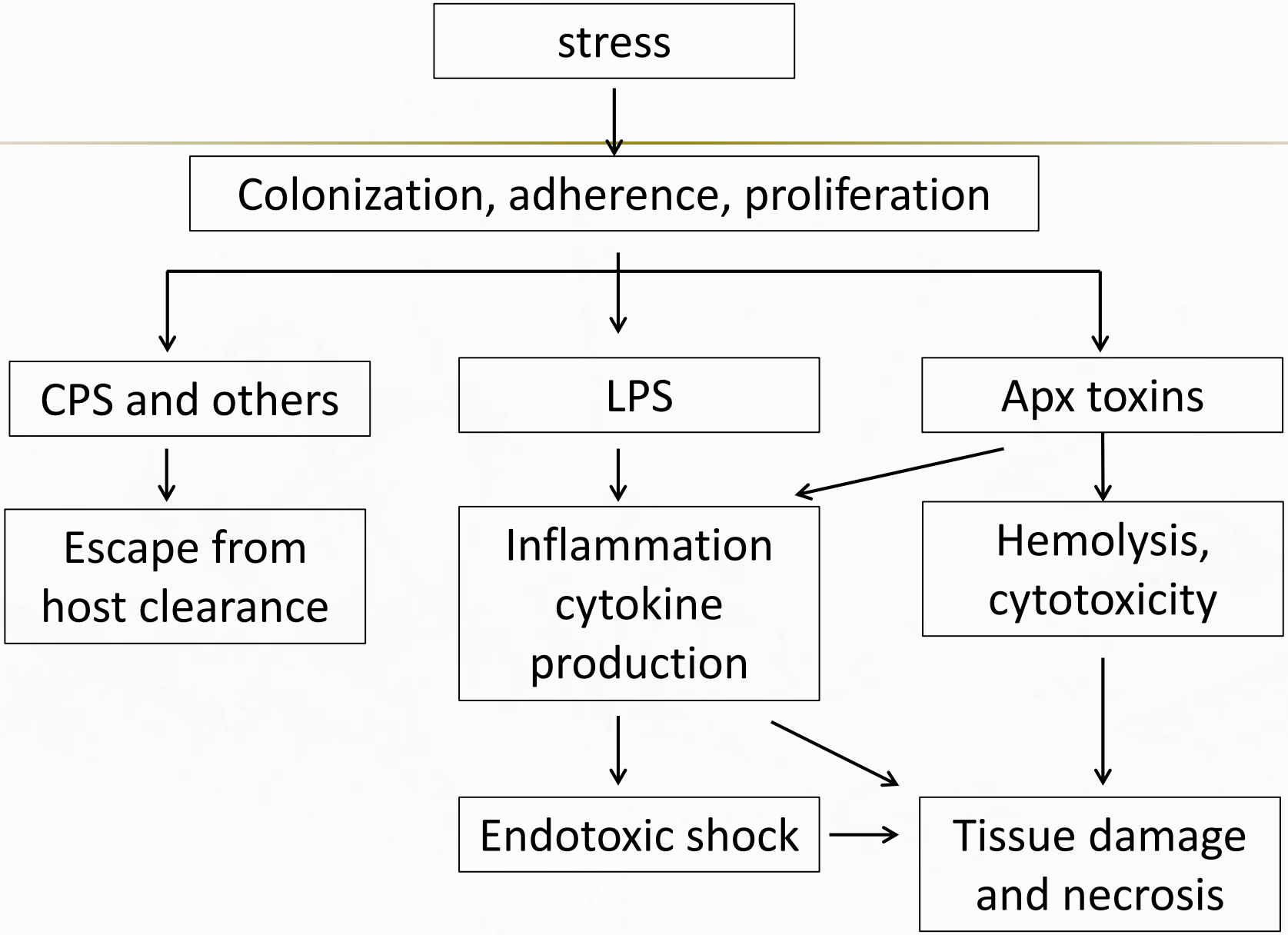
3. Avoidance of host clearance mechanisms: capsule / LPS, resistance to phagocytic killing, oxidative-stress defense

4. Damage to host tissues: Apx toxin, LPS

APX toxin: major virulence determinants

	activity		serotype
	hemolytic	cytotoxic	
ApxI	Strong	Strong	1, 5, 9, 10, 11, 14
ApxII	Weak	Moderate	All but 10, 14
ApxIII	None	Strong	2, 3, 4, 6, 8, 15
ApxIV	weak	ND	all

Bosse et al., 2002*, Microbe. Infect.* 4:225-235


stress
Colonization, adherence, proliferation
CPS and others
LPS
Apx toxins
Escape from host clearance
Inflammation cytokine production
Hemolysis, cytotoxicity
Endotoxic shock
Tissue damage and necrosis

臨床症狀 (clinical signs)

Clinical signs vary with the age of the animals, state of immunity, environmental conditions, and degree of exposure to the infectious agent.

- 甚急型 (Peracute form)
  - Suddenly ill, fever (to 41.5°C), apathy, anorexia, short period of diarrhea and vomiting.
    - Affected animals lie on floor without distinct respiratory signs.
    - Cyanosis: nose, ears, legs, and later the whole body.
    - terminal phase
      - Severe dyspnea with open-mouth breathing, remain in a sitting posture
      - Foamy, blood-stained nasal and oral discharge (口鼻泡沫血樣液體流出)
    - Death occurs within 24-36 h of development of clinical signs.
    - Some pigs may die suddenly without clinical signs.

臨床症狀 (clinical signs)

* 急性型 (Acute form)
  - Fever (40.5-41C), skin may be reddened
  - Depressed, reluctant to rise, refuse food and drink
  - Severe respiratory symptoms: dyspnea, cough, mouth breathing
  - Congestion of the extremities
  - Marked loss of condition within 24 h of onset

臨床症狀 (clinical signs)

- 亞急性及慢性型 (subacute and chronic form)
  - Develop after the disappearance of acute signs
  - Little or no fever, spontaneous or intermittent cough
  - Reduced appetite, intolerance of exercise
  - Exacerbated by co-infections (mycoplasmal, bacterial, or viral)
- 無臨床症狀型 (subclinical form)
  - With no apparent clinical signs and no significant lesions at slaughter

高熱(體表潮紅)、四肢耳翼發紺、呼吸困難(張口呼吸)等







感染豬隻呈現精神沈鬱，嚴重者口鼻部位有鮮血流出、死亡







宣詩玲 國立中興大學

豬病診療 楊平政等 1996 台灣養豬科學研究所

急性死亡豬隻口鼻部位泡沫樣鮮血流出





Figure 48.1 Animal that died from peracute pleuropneumonia caused by *Actinobacillus pleuropneumoniae* presenting with typical foamy blood-tinged nasal discharge. *Source:* Courtesy of Dr. Enric Marco.



邱慧英 國立中興大學

病變 (lesions)

- Gross

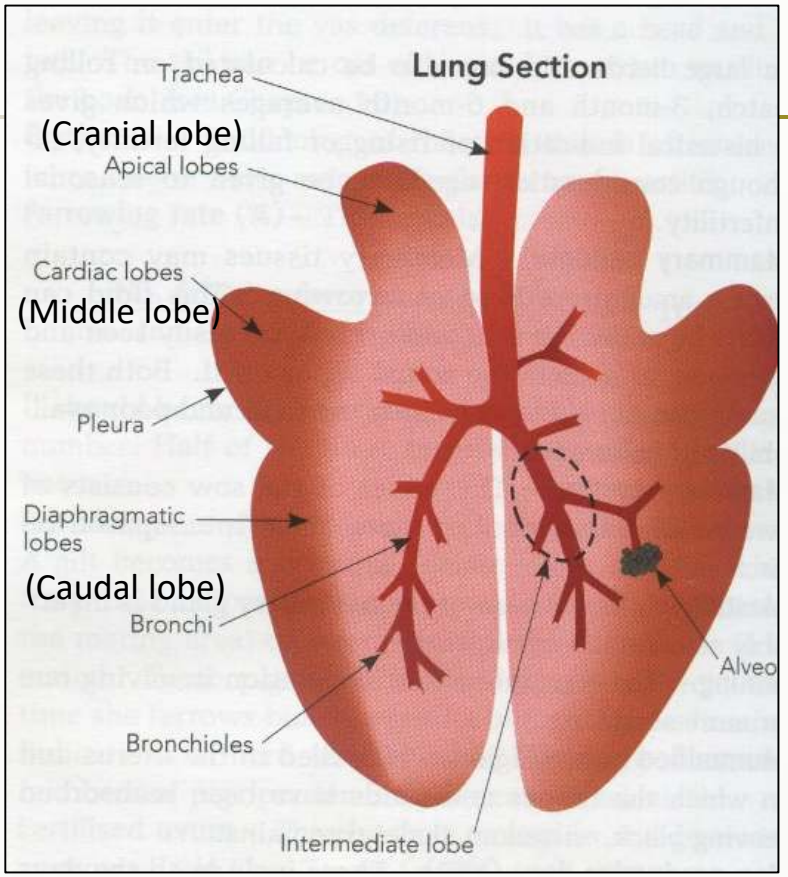
  Located mainly in the respiratory tract

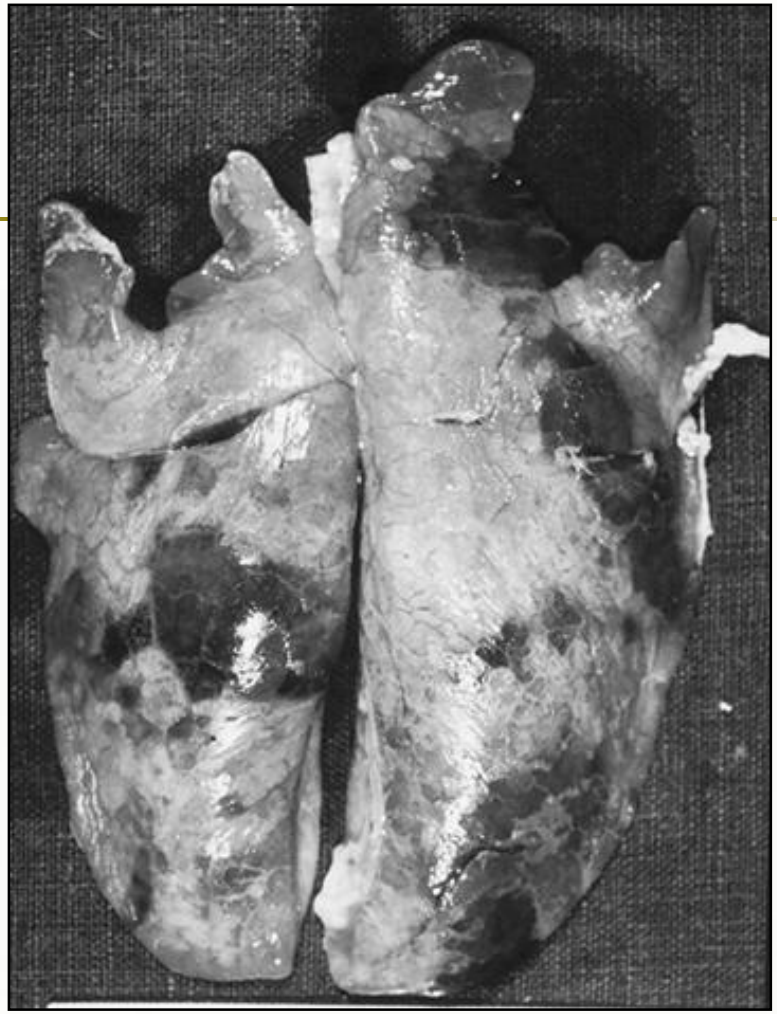
  the pneumonia is most bilateral, with involvement of the cardiac and apical lobes, and part of the diaphragmatic lobes. Pneumonic lesions are often local and well demarcated.

  trachea and bronchi are filled with a foamy, blood-tinged, mucous exudate

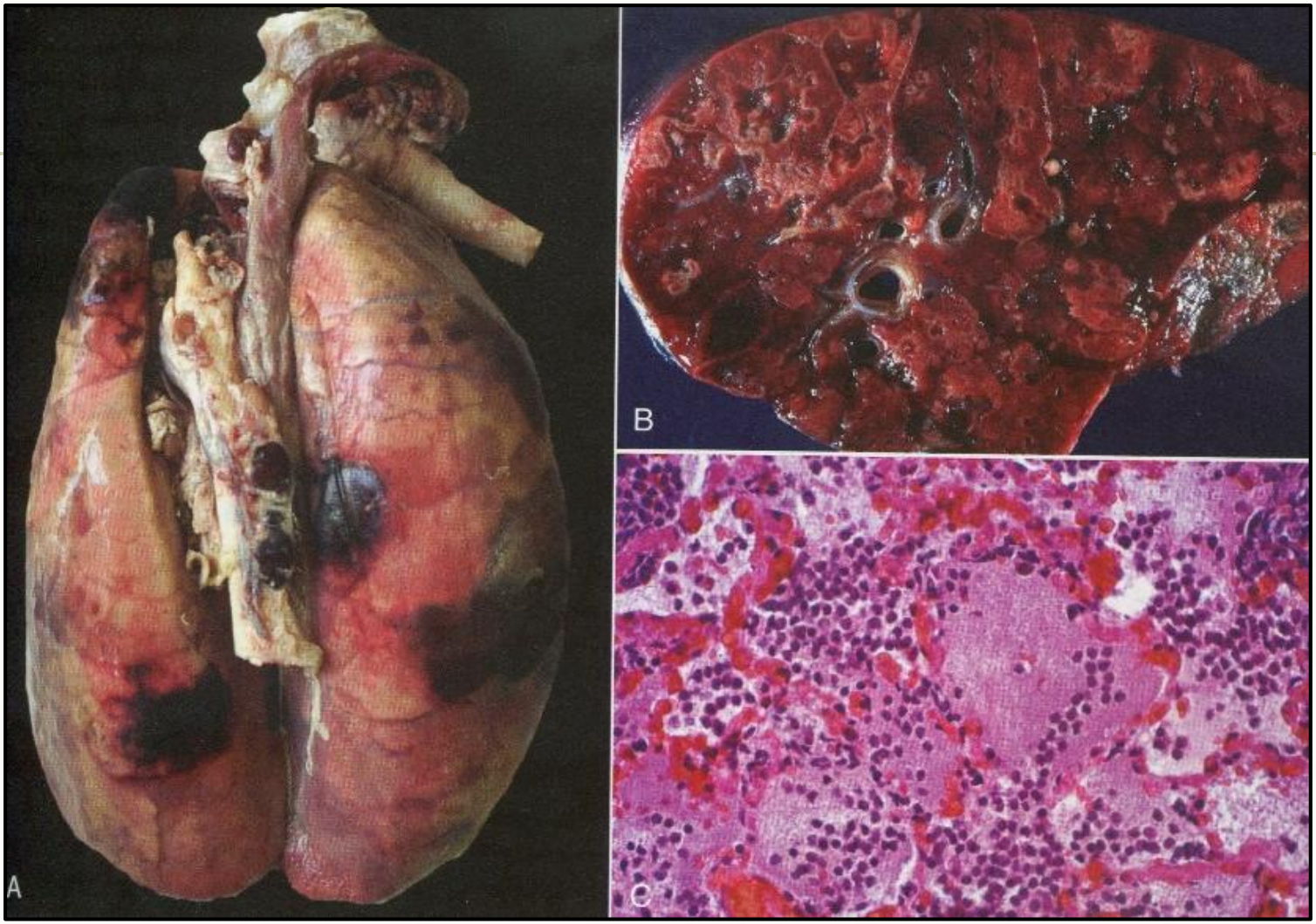
  Thoracic cavity contains a blood-tinged fluid

  Fibrinous pleurisy [=pleuritis] (纖維素性胸膜炎) : lungs adhere to parietal pleura

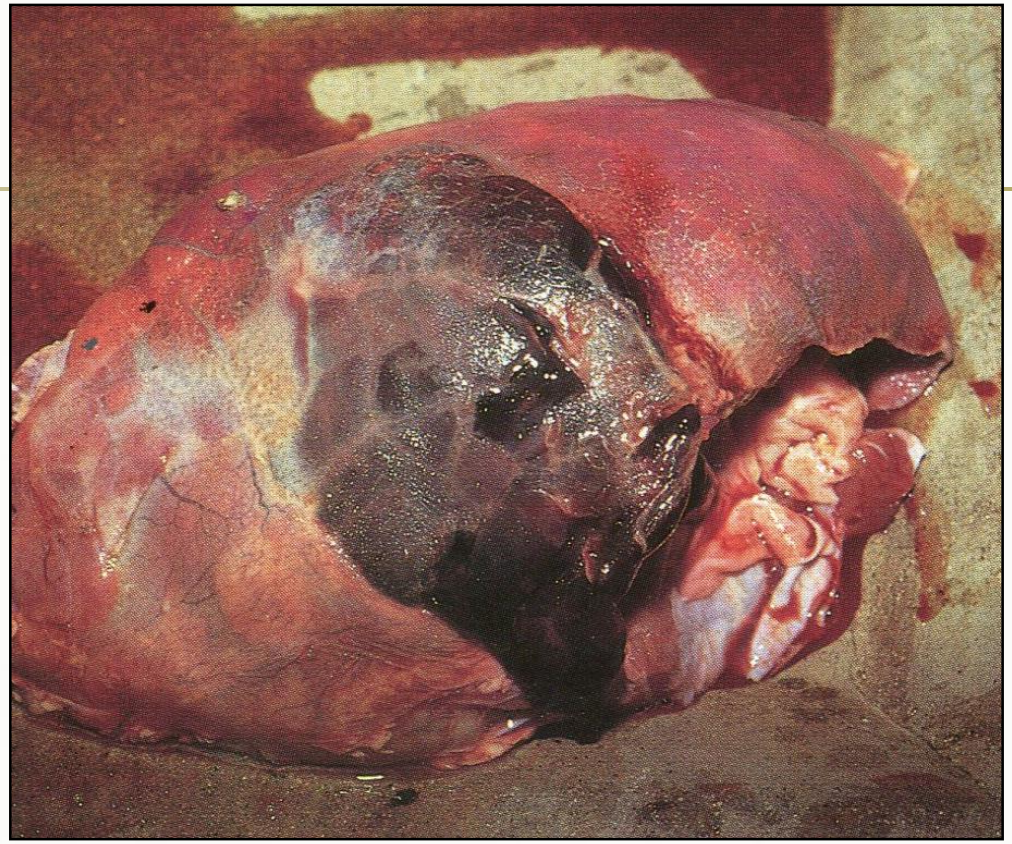




Well-circumscribed pneumonic area in different lobes with fibrinous pleurity
From *Disease of Swine*

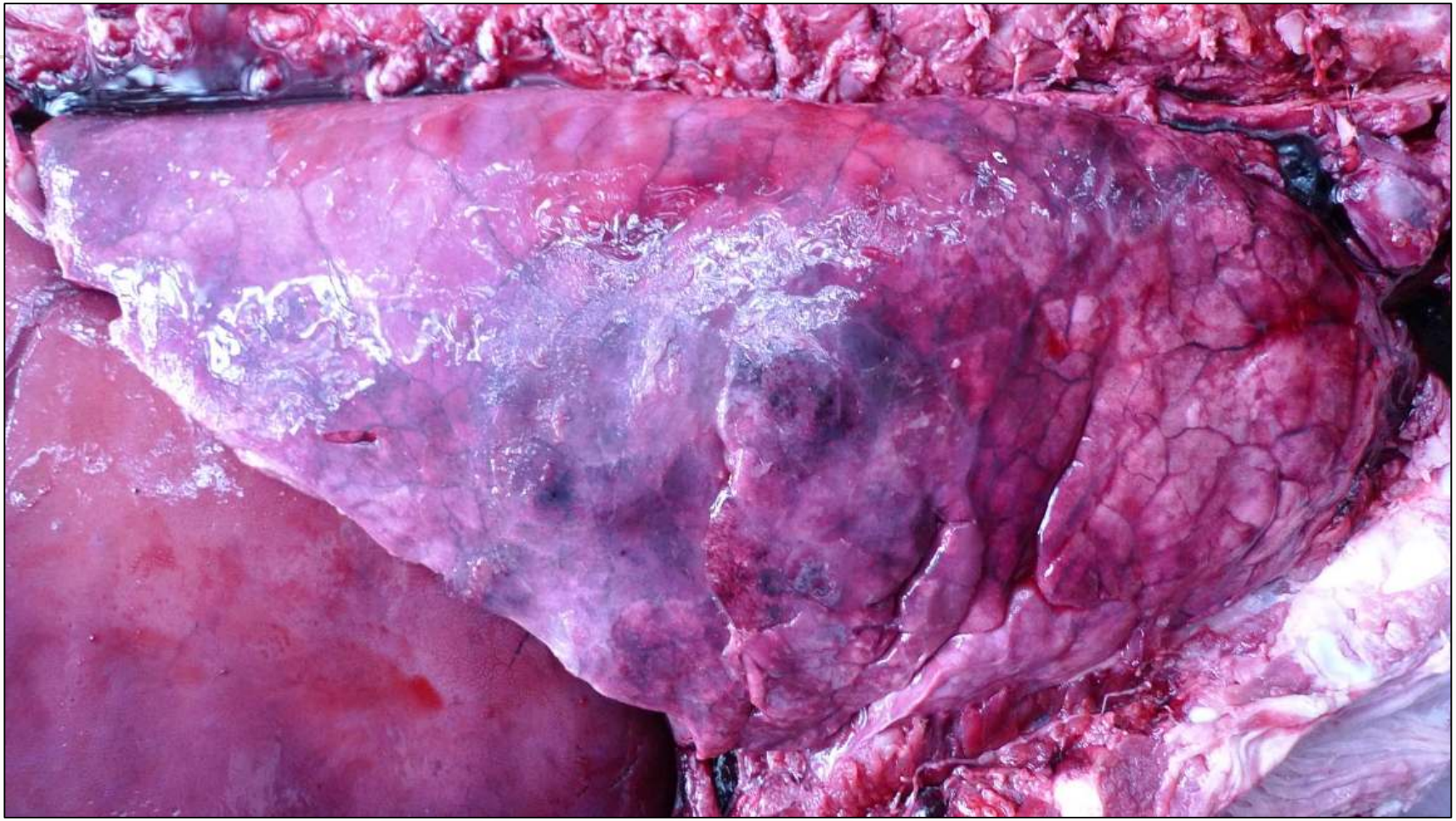


Zachary, Pathologic Basis of Veterinary Disease 6th ed., 2017, Elsevier



A typical well-demarcated pneumonic lesion can be seen. The surfaces of the lesion and the adjacent lung are covered by fibrin which in older lesions becomes organized into fibrous pleurisy.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



邱慧英 國立中興大學

診斷 (diagnosis)

* Pathological

Post-mortem

1. acute, fibrinous, hemorrhagic pleuropneumonia

2. the lung is dark red and firm to touch, with blood running from the cut surfaces

3. lesions are black-red raised areas, especially in dorsal parts of both lungs

4. many cases also have fibrinous pleurisy and pericarditis

- Microbiological
  - From freshly dead animals
  - Blood agar supplemented with NAD or chocolate agar
  - CAMP phenomenon, urease activity test
- Polymerase chain reaction (PCR)
- Serological test
  - ELISA

治療 (treatment)

Acute cases require early parenteral antimicrobial therapy.

Antimicrobial selection should preferably be based on culture and susceptibility testing.

Severely affected pigs may die before antimicrobial treatment becomes effective.

Herd-level outbreak control requires simultaneous improvement of management and environmental conditions.

Antimicrobial resistance should be considered when selecting therapy.

預防 (Prevention)

- Biosecurity / prevent introduction of carriers
- Reduce animal mixing and environmental stress
- All in/all out (統進統出) system in fattening unit
- Vaccination
  - Killed organisms (bacterin)
  - Toxin subunit vaccine
  - Live vaccine

114年度國考試題

下列有關豬放線桿菌胸膜肺炎之敘述，何者錯誤？

A. 只發生於肥育期豬隻

B. 造成出血性壞死性肺炎

C. 病變主要由Apx毒素及LPS傷害所致

D. 本病通常不感染人類

110年度國考試題

下圖為一隻肥育豬隻，箭頭指示處顯示此斃死豬隻口鼻有泡沫樣血液流出。此豬隻死亡前有咳嗽、呼吸困難、發紺等臨床症狀，剖檢可見肺臟有水腫及出血。若從肺臟病灶分離出一種革蘭氏陰性細菌，CAMP test為陽性，其最有可能是感染下列那一個病原？



A. *Pasteurella multocida*

B. *Actinobacillus pleuropneumoniae*

C. *Yersinia pestis*

D. *Escherichia col*

109年度國考試題

下列何者主要感染生長肥育豬隻，發生甚急性型感染時，豬隻呈現口鼻出血，並於短時間內死亡，剖檢以出血性或壞死性纖維素性胸膜肺炎為主要特徵？

A. *Mycoplasma hyopneumoniae*

B. *Actinobacillus pleuropneumoniae*

C. *Streptococcus suis*

D. *Brachyspira hyodysenteriae*

105年度國考試題

下列有關*Actinobacillus pleuropneumoniae* RTX毒素（repeats in the structural toxin）之描述，何者錯誤？

A. 具有ApxI，ApxII，ApxIII與ApxIV四種

B. ApxIII具有較強的溶血性

C. ApxII具有弱溶血性

D. ApxI，ApxII，ApxIII均具有細胞毒性

105年度國考試題

下列那一種豬隻病原不會感染人？

A. *Salmonella* Choleraesuis

B. *Brucella suis*

C. *Actinobacillus pleuropneumoniae*

D. *Erysipelothrix rhusiopathiae*

104年度國考試題

* 下圖所示為位於山坡地區某一貫式豬場猝死肥育豬之症狀及病變，發病率約6－9%，最有可能之疾病為何？





A.豬瘟
B.豬萎縮性鼻炎
C.豬放線桿菌胸膜肺炎
D.豬鏈漿菌症

103年度國考試題

* 胸膜肺炎放線桿菌（*Actinobacillus pleuropneumoniae*）所分泌的APX毒素，其作用機制最接近下列何者？

A. *Escherichia coli*的heat-labile enterotoxin

B. *Pseudomonas aeruginosa*的exotoxin A

C. *Clostridium tetani*的tetanus toxin

D. *Escherichia coli*的HlyA toxin

103年度國考試題

* 下列有關豬放線桿菌胸膜肺炎之敘述，何者正確？
  A. 單側性肺炎是主要的剖檢病灶
  B. 培養該病原菌時需添加nicotinamide adenine dinucleotide (NAD)
  C. 臺灣經常爆發第四血清型
  D. 感染豬齡以哺乳豬最常見

101年度國考試題

* 豬罹患放線桿菌胸膜肺炎，主要臨床症狀為下列何者？

A.急促腹式呼吸

B.神經症狀

C.下痢

D.嘔吐、食慾不振

100年度國考試題

* 下列有關豬罹患放線桿菌胸膜肺炎之敘述，何者正確？

A.剖檢可見大腸鈕扣狀潰瘍

B.肥育豬感染最為普遍

C.是造成哺乳豬下痢的主因

D.是造成氣腫疽的主因

97年度國考試題

* 自豬肺臟病灶分離*Actinobacillus pleuropneumoniae*所使用的培養基須添加下列何種成分？

A. lysine

B. neuraminidase

C. nicotinamide adenine dinucleotide

D. erythromycin

97年度國考試題

* 下列有關豬放線桿菌胸膜肺炎急性病例之敘述，何者正確?

A. 臨床上可見口鼻滲血

B. 哺乳豬的發生率最高

C. 病原菌屬於革蘭氏陽性菌

D. 病原菌屬於抗酸菌

96年度國考試題

* 豬感染下列何種疾病，在臨床上可見口鼻出血？

A. 豬流行性感冒

B. 黴漿菌病

C. 巴氏桿菌病

D. 放線桿菌胸膜肺炎

萎縮性鼻炎(atrophic rhinitis)

- Atrophic rhinitis (or infectious atrophic rhinitis) has been recognized for nearly two centuries.
- Growth rate can be reduced, and the predisposition to pneumonia is increased.
- An infected herd may show evidence of snout deformity (鼻吻部變形) in abattoir surveys.

病因 (Etiology)

- nonprogressive atrophic rhinitis (NPAR)

  --支氣管敗血性博德氏菌 (*Bordetella bronchiseptica*)

- progressive atrophic rhinitis (PAR)

  --產毒型多殺性巴斯德桿菌 (toxgenic *Pasteurella multocida*) or other agents

病因 (Etiology)

	NPAR	PAR
Major agent	*B. bronchiseptica*	Toxgenic *P. multocida*
Severity	mild/transient	severe/progressive
Turbinate damage	reversible/mild	permanent/severe
Growth effect	limited	significant

病因 (Etiology)

- Among *Bordetella*, *B. bronchiseptica* is the only species of important in swine.

- *B. bronchiseptica* infects a broad range of mammals, occasionally causing acute disease, but more often resulting in chronic or inapparent infection of the upper respiratory tract.



鼻吻部彎曲





肥育豬，鼻子歪向右邊，上顎比下顎短。

陳世平主編屠宰豬隻肉眼病理學圖譜2008年動植物防疫檢疫局 台灣動物科技研究所



Brachygnathia superior 上顎短顎症



Deviation of the snout

Handbook of Pig Medicine by Jackson and Cockcroft , 2007, Elsevier



Secretion or bleeding from the nose

豬病診療 楊平政等 **1996**
台灣養豬科學研究所

公共衛生學 (public health)

- Human *B. bronchiseptica* infection is uncommon and occurs mainly in immunocompromised individuals.

流行病學 (Epidemiology)

- Transmission of *B. bronchiseptica* occurs primarily by aerosol droplets. Spread by direct or droplet contact between pigs. The main route of spread is between an infected sow and her litter in their first week of life.

- Further spread occurs when batches of piglets are mixed.

流行病學 (Epidemiology)

- *B. bronchiseptica* has been isolated from rodents, birds, raccoons, opossums, and other animals trapped in close proximity to swine.

- Transmission via contaminated fomites should not be excluded since *B. bronchiseptica* remains viable in soil (up to 45 days), or lake water.

致病機制 (Pathogenesis)

- The pathogenesis of *B. bronchiseptica* is dependent on the sequential, coordinated synthesis of an array of virulence factors.

- Expression of most virulent genes requires co-expression of the BvgAS system.

致病機制 (Pathogenesis)

Virulence factor of *Bordetella bronchiseptica*

Virulence factor	**activity**
Filamentous haemagglutinin	Bind to cilia
pertactin	Bind to cells
fimbriae	Mediate attachment to cells
Adenylate cyclase-haemolysin	Interfere with phagocytic cell function
Tracheal cytotoxin	Inhibits ciliary action, kill ciliated cells
Dermonecrotoxic toxin	Induce skin necrosis, impairs osteogenesis
osteotoxin	Toxic for osteoblasts
lipopolysaccharide	Stimulate cytokine release

Veterinary Microbiology and Microbial Disease, Quinn et al., 2002, Blackwell Science

* Dermonecrotic toxin (DNT)

exert pleiotropic effects, including impairment of bone formation.

strains with reduced dermonecrotic activity failed to induce turbinate and lung lesions.

Other studies also indicated that the toxin is essential for the development of pneumonia and turbinate atrophy in both mice and pigs.

Pasteurella multocida

- *Pasteurella multocida* toxin (PMT)

  --PMT is an important virulence factor responsible for AR in pigs.

  --a 146-kDa protein encoded on a lysogenic bacteriophage residing only in the genome of the toxing-producing strain of *P. multocida*

  PMT alters bone remodeling → reduced bone formation and increased bone resorption → progressive turbinate atrophy.

- Coinfection with *B. bronchiseptica* and other pathogens in the respiratory tract affects disease severity.

- The ability of *B. bronchiseptica* to predispose to colonization of the upper respiratory tract with *P. multocida* leading to PAR has been well established.

- It was suggested that type D capsule of *P. multocida* could potentially interact with a heparin-binding domain of the FHA of *B. bronchiseptica*, using it as a bridge to mediate binding to cells of the respiratory tract.

臨床症狀 (Clinical Signs)

Bordetella bronchiseptica (NPAR)

Rhinitis

Piglets with 3-4 weeks of age, sneeze, snuffle, and snort with a variable degree of catarrhal rhinitis. Serous or mucopurulent nasal discharges.

Brochopneumonia

usually seen in very young pigs (3-4 days).

臨床症狀 (Clinical Signs)

Pasteurella multocida (PAR)

Piglets with 4-12 weeks of age

Sneezing and snuffling

Nose bleeding, snout deformation

Growth retardation

Turbinate atrophy

Septum deviation, conspicuous deformities of face



患病豬隻可見打噴嚏、鼻吻部彎曲，眼窩下有污痕，甚至有流鼻血等症狀。



患病豬隻生長受抑制，與同年齡的豬隻比較，體型明顯較瘦小。

張聰洲等 經濟動物常見疾病-豬隻篇 2015

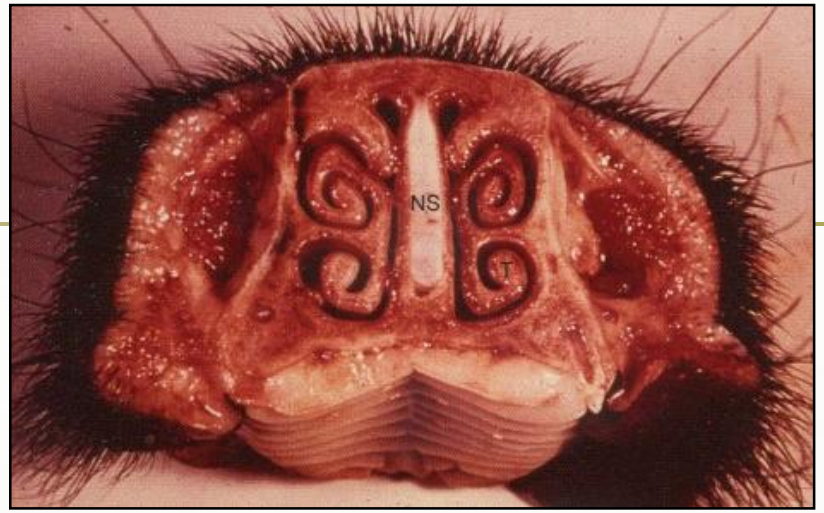
病變 (lesions)

Restricted to nasal cavity and skull

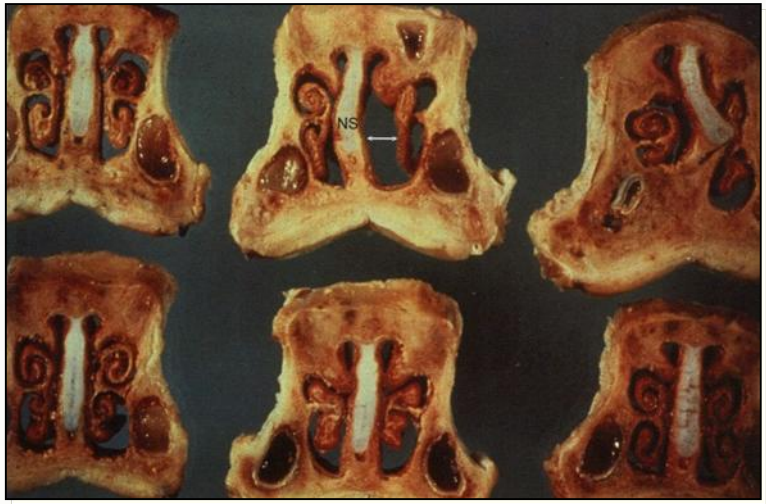
Atrophy of the ventral and dorsal turbinates

In severe cases: atrophy of ethmoidal turbinates may occur

Complete absence of all turbinates structures seen in the most severe form



Nasal turbinates from a normal pig. The nasal septum (NS) is straight and the turbinate (T) fill the nasal cavity



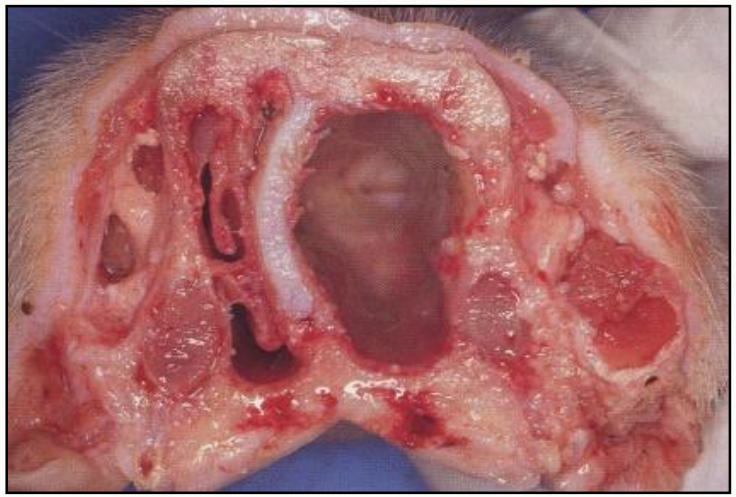
Gross lesions of atrophic rhinitis in swine. The nasal septum (NS) is distorted and turbinates are shrunken and misshapen resulting in increased airspace (arrow) in the nasal cavity



Early turbinate damage

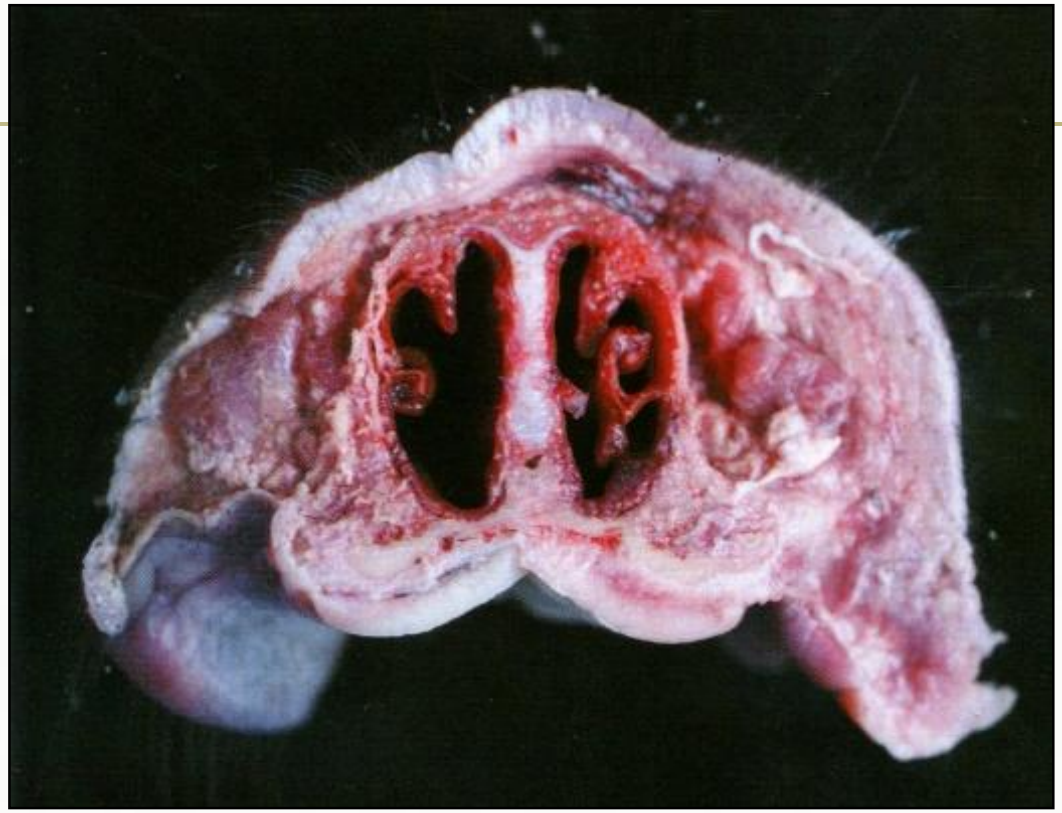


bilateral turbinate destruction



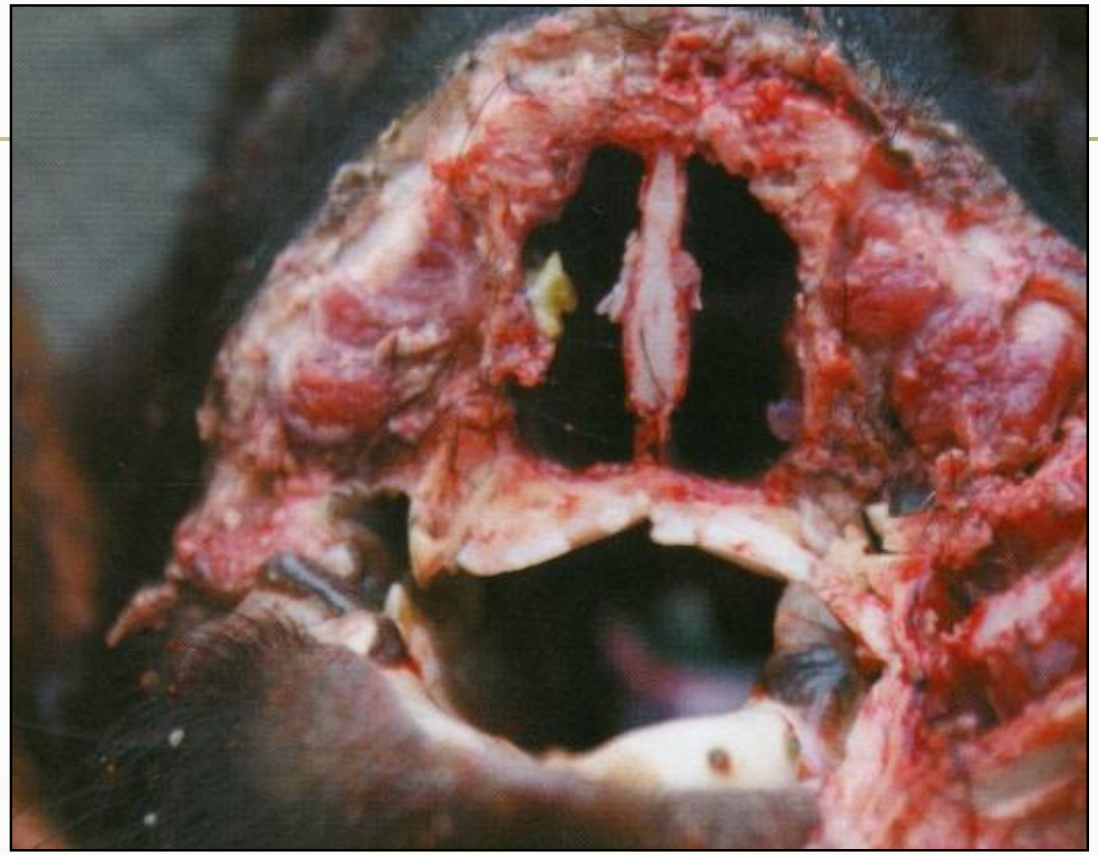
Unilateral turbinate destruction

Handbook of Pig Medicine by Jackson and Cockcroft , 2007, Elsevier



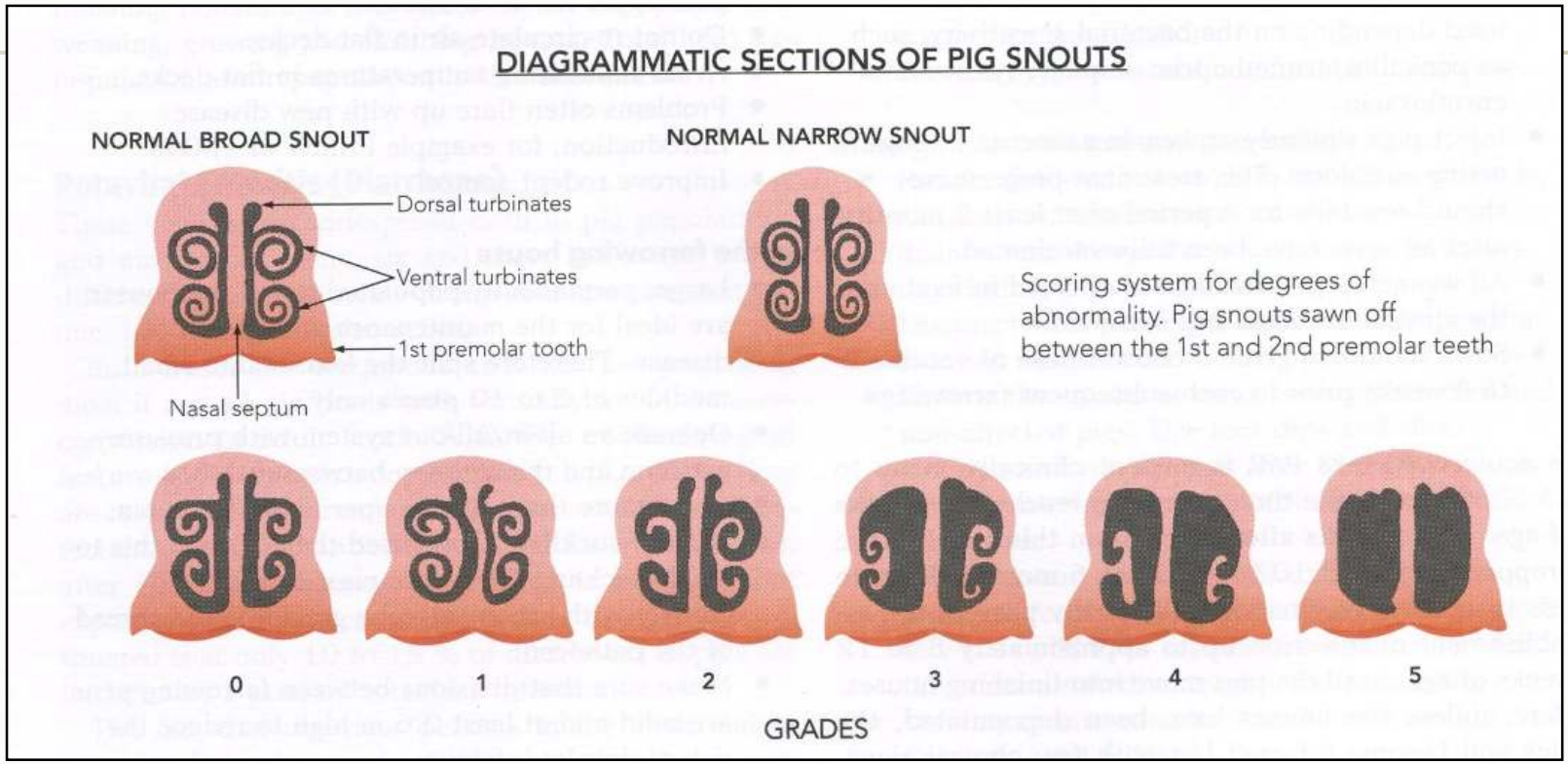
鼻中膈正常，右邊鼻甲介骨呈中度病變，左邊鼻甲介骨幾乎萎縮消失。

陳世平主編屠宰豬隻肉眼病理學圖譜2008年動植物防疫檢疫局 台灣動物科技研究所



鼻中膈變薄，兩邊的鼻甲介骨已萎縮消失呈現空腔。

陳世平主編屠宰豬隻肉眼病理學圖譜2008年動植物防疫檢疫局 台灣動物科技研究所



Muirhead and Alexander, Managing Pig Health 2nd ed., 2013

診斷 (diagnosis)

- Clinical diagnosis
- Postmortem
- Cultural diagnosis (Detection of toxigenic *P. multocida* / toxin-related genes is important for confirmation of progressive atrophic rhinitis)

治療與預防 (treatment and prevention)

vaccination of sows

colostral immunity

improved hygiene

reduced transmission

herd replacement/biosecurity

antimicrobial treatment only when clinically justified

治療與預防 (treatment and prevention)

- Vaccination of sows around 6 weeks and again around 2 weeks prior to farrowing appears to work well to protect piglets and minimize transmission.

- Although vaccine may prevent clinical outbreaks and decrease *Bordetella* burden, they are highly unlikely to prevent colonization.

115年度國考試題

下列何種病原所分泌的毒素為豬萎縮性鼻炎最主要之致病因子？

A. *Bordetella bronchiseptica*

B. *Staphylococcus aureus*

C. *Pasteurella multocida*

D. *Clostridium perfringens*

114年度國考試題

下列何者不是*Bordetella bronchiseptica* 造成豬隻呼吸道感染的致病因子？

A. filamentous hemagglutinin（FHA）

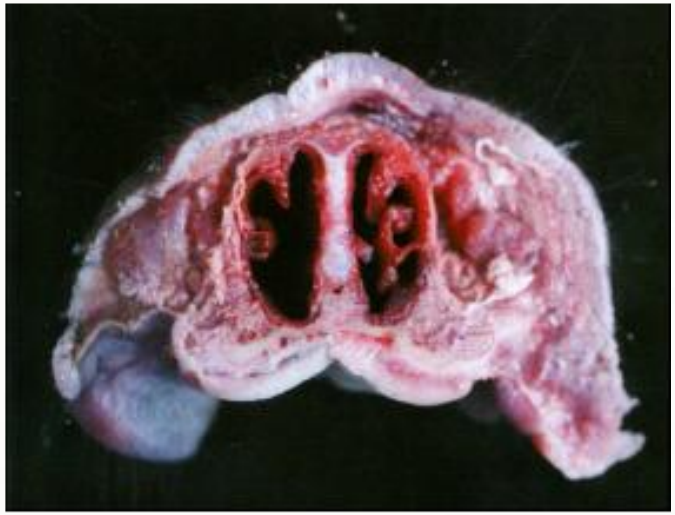
B. dermonecrotic toxin（DNT）

C. tracheal cytotoxin（TCT）

D. exfoliative toxins（EFT）

113年度國家考試題目

下圖為豬隻傳染病引起之病變，推測致病原因是*Bordetella bronchiseptica*，其與下列何病原的感染最為相關？



A. *Salmonella* Typhimurium

B. *Escherichia coli*

C. *Pasteurella multocida*

D. *Staphylococcus aureus*

111年度國考試題

下列有關豬萎縮性鼻炎之敘述，何者正確？

A. 因雙側鼻甲骨萎縮導致豬隻鼻吻部變短

B. 患病豬隻眼窩下方污痕是因結膜炎所導致

C. 於豬隻第一前臼齒處橫切面為觀察鼻甲骨萎縮最理想部位

D. 本病尚無疫苗可用，防治策略以抗生素治療為主

105年度國考試題

豬隻常見且易造成肺炎及萎縮性鼻炎之*Pasteurella multocida*血清型為何？

A. A，B

B. A，D

C. A，E

D. B，F

104年度國考試題

下列有關*Bordetella bronchiseptica* 的敘述，何者錯誤？

A. 該菌為具運動性之革蘭氏陰性短桿菌

B. 與*Pasteurella multocida*可引起萎縮性鼻炎

C. 毒力株具有凝集血球能力

D. 無法在MacConkey 培養基上生長

104年度國考試題

*Pasteurella multocida*依照細菌何種物質進行血清型分型為A、B、D、E與F等五種？

A. capsular polysaccharides

B. flagellum

C. pilus

D. peptidoglycan

102年度國考試題

*Pasteurella multocida*與豬隻罹患下列何種疾病最具相關性？

A. 多發性漿膜炎

B. 萎縮性鼻炎

C. 新生仔豬下痢

D. 水腫病

98年度國考試題

*Bordetella bronchiseptica*主要引起豬隻何種疾病？

A.萎縮性鼻炎

B. 腸炎

C. 心肌炎

D. 腦膜炎

96年度國考試題

豬萎縮性鼻炎的主要症狀為下列何者？

A. 迴旋、顫抖

B. 打噴嚏、聳鼻

C. 嘔吐、下痢

D. 發燒、呼吸困難

副豬格拉氏菌

(*Glaesserella parasuis;* formerly *Haemophilus parasuis*)

關聯 (Relevance)

* Glässer於1910年首先從具有纖維素性多發性漿膜炎 (fibrinous polyserositis) 的豬隻滲出液分離出一種桿菌，多年後才確認此細菌為*Glaesserella parasuis*。

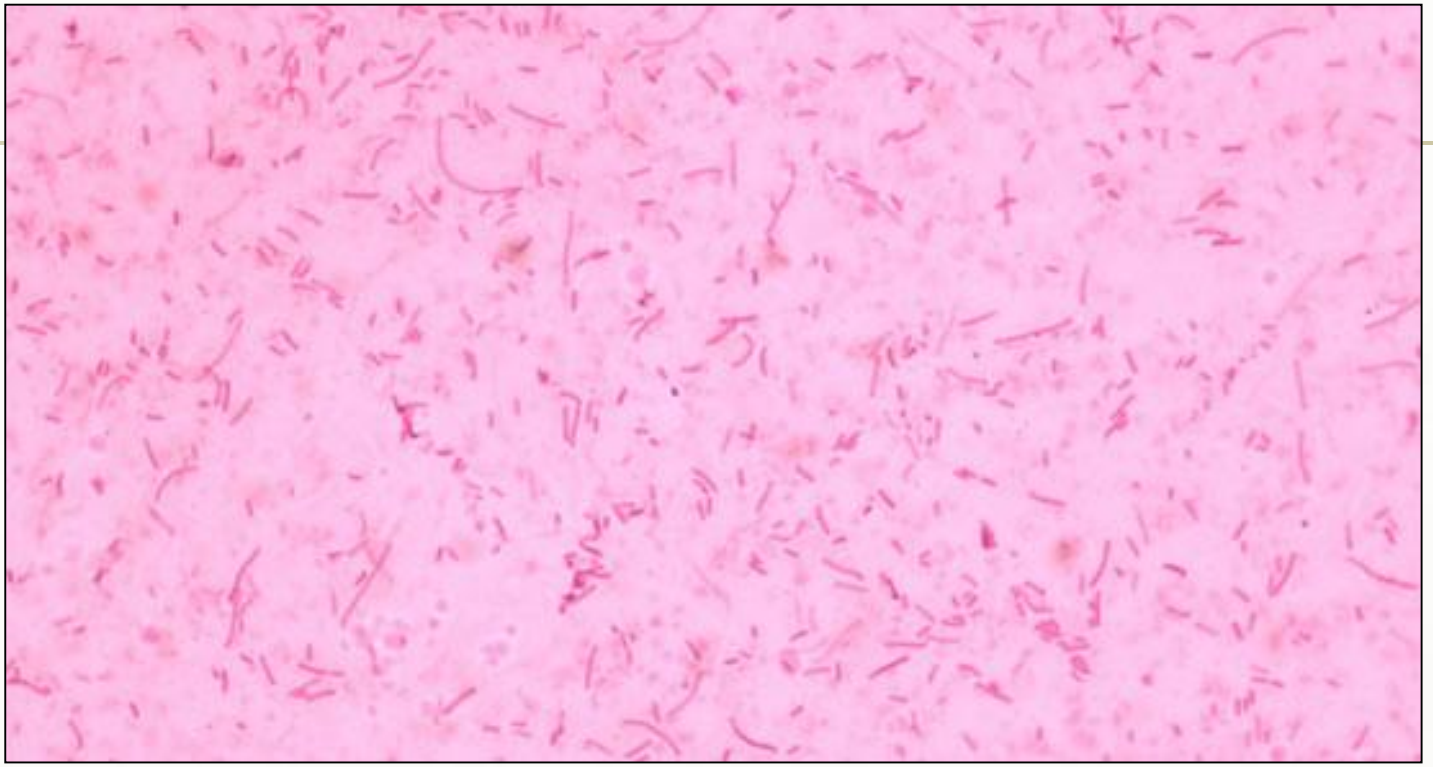
* 造成豬隻的多發性漿膜炎、關節炎、或腦膜炎，稱之為Glässer disease。

* Outbreak may be linked to stress-associated events including coinfections and moving and commingling animals.

* One of the main infectious problems in the nursery, also affecting growing pigs and sows.

病因 (Etiology)

- *G. parasuis* is a Gram negative bacterium.
- *G. parasuis* is similar to *H. influenzae*, but of porcine origin.
- *G. parasuis* does not require X factor (hemin) to grow, however, it requires V factor (NAD).
- *G. parasuis* isolates can be classified into 15 serovar groups.



Gram stain of *Glaeserella parasuis*. Short gram negative bacilli are observed and others that are longer (pleomorphic 多型性)

http://www.pig333.com/glasser_disease/laboratory-diagnosis_1590/



Growth of *Glaeserella parasuis* in pure cultivation on a plate of chocolate agar

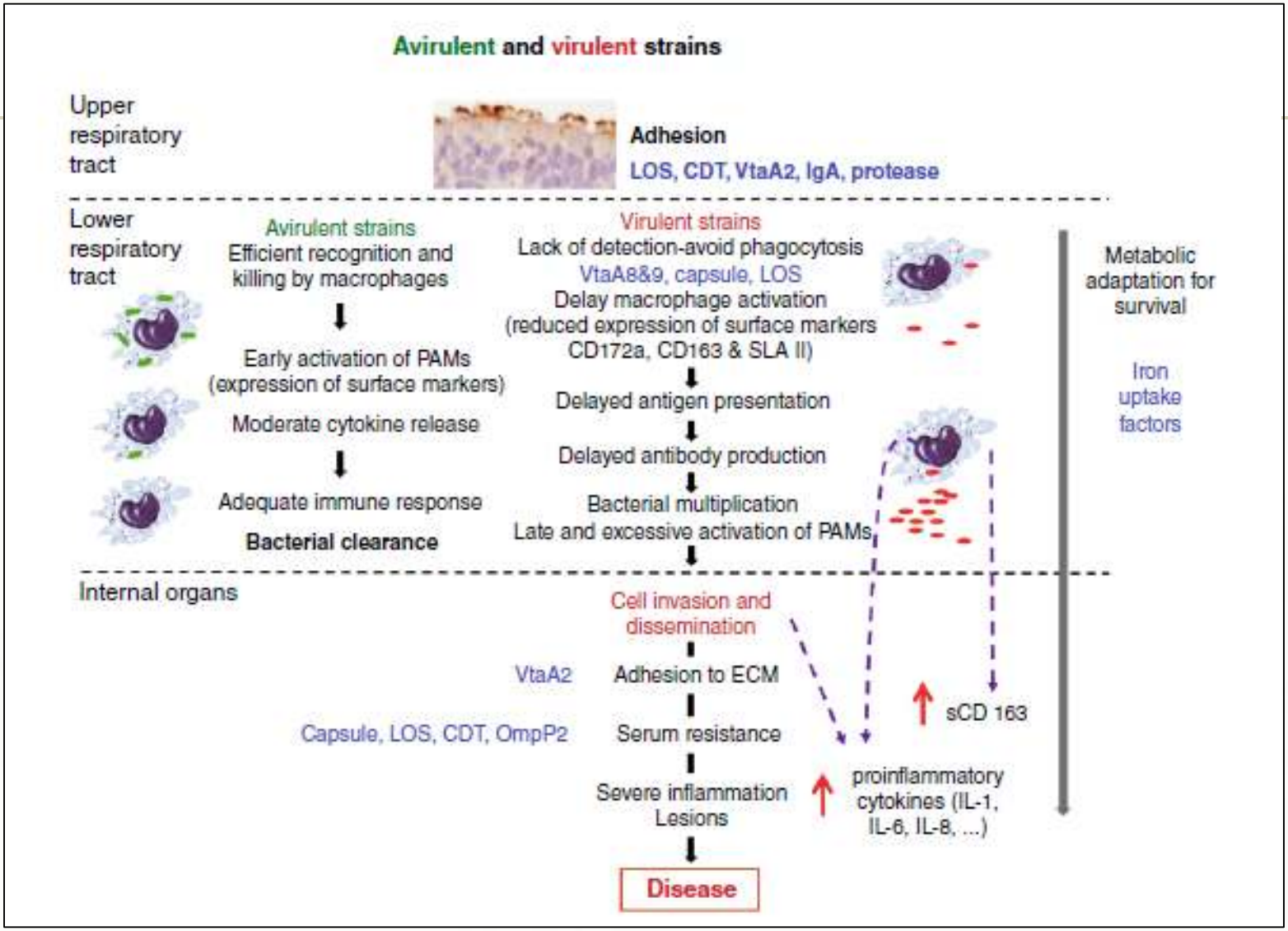
http://www.pig333.com/glasser_disease/laboratory-diagnosis_1590/

流行病學 (Epidemiology)

- *G. parasuis* is found in the nasal cavity of many pigs. It is a member of the normal respiratory microbiota and is ubiquitous in swine herd.
- The onset of disease may be associated with exposure to stress such as transport or movement to new pens, unstable temperature, poor ventilation.
- Domestic pig and wild boar are the only known hosts for this bacterium.
- Transmission of Glässer's disease occurs through contact of carrier or diseased pigs with susceptible animals.

致病機制(Pathogenesis)

- Adherence, biofilm formation
- Phagocytosis resistance
- *H. parasuis* invades endothelial cells and induce apoptosis and production of proinflammatory IL-6 and IL-8
- respiratory colonization → systemic invasion → bacteremia → serosal inflammation → fibrin deposition.


Avirulent and virulent strains
Upper respiratory tract
Adhesion
LOS, CDT, VtaA2, IgA, protease
Lower respiratory tract
Avirulent strains
Efficient recognition and killing by macrophages
Early activation of PAMs (expression of surface markers)
Moderate cytokine release
Adequate immune response
Bacterial clearance
Virulent strains
Lack of detection-avoid phagocytosis
VtaA8&9, capsule, LOS
Delay macrophage activation
(reduced expression of surface markers CD172a, CD163 & SLA II)
Delayed antigen presentation
Delayed antibody production
Bacterial multiplication
Late and excessive activation of PAMs
Metabolic adaptation for survival
Iron uptake factors
Internal organs
Cell invasion and dissemination
VtaA2
Adhesion to ECM
Capsule, LOS, CDT, OmpP2
Serum resistance
sCD 163
Severe inflammation Lesions
proinflammatory cytokines (IL-1, IL-6, IL-8, ...)
Disease

臨床症狀(Clinical signs)

* Clinical signs are mainly observed in 4 to 8-week-old pigs.

* Peracute disease has a short course (48 hours) and may result in sudden death without characteristic gross lesions.

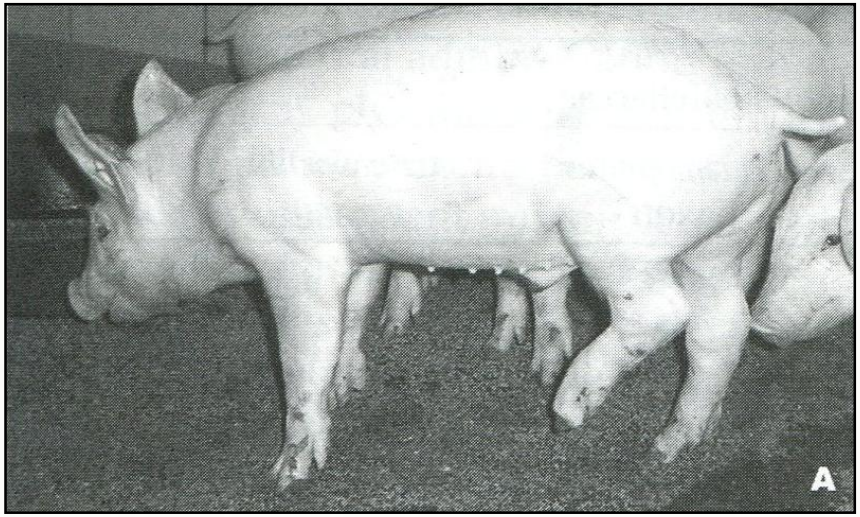
* The typical clinical signs of acute disease include high fever (41.5^{o}C), coughing, abdominal breathing, lameness, with swollen joints and pigs walking with short strides and on tiptoes (利用腳尖小步行走), and central nervous signs such as lateral decubitus, paddling, and trembling.

病變 (Lesions)

- Peracute: die without characteristic gross lesions, but may show petechial hemorrhages in some tissues.

- Acute: fibrinous or fibrinopurulent polyserositis (纖維化膿性多發性漿膜炎), polyarthritis, and meningitis. The fibrinous exudate can be observed on the pleura, pericardium, peritoneum, synovia, and meninges, and is usually accompanied by an increased amount of fluid.

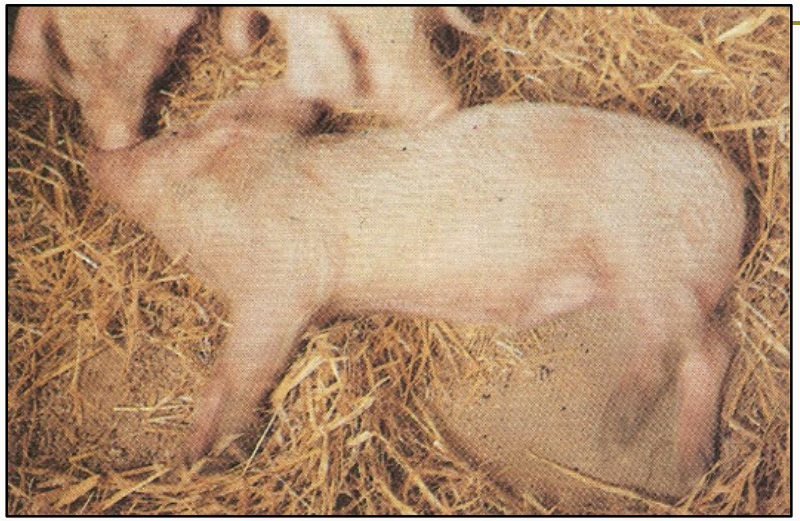
- Lack of characteristic gross lesions is also common in a number of pigs showing neurological clinical signs.





lameness

Left: Diseases of Swine; right: 豬病診療
楊平政等 1996 台灣養豬科學研究所



Glässer's disease. A pig with meningitis. Note the evidence of paddling movements with its limbs.



Glässer's disease. Carcass of a pig which has died from Glässer's disease. Note the cyanosis of the extremities and the focal skin lesions.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

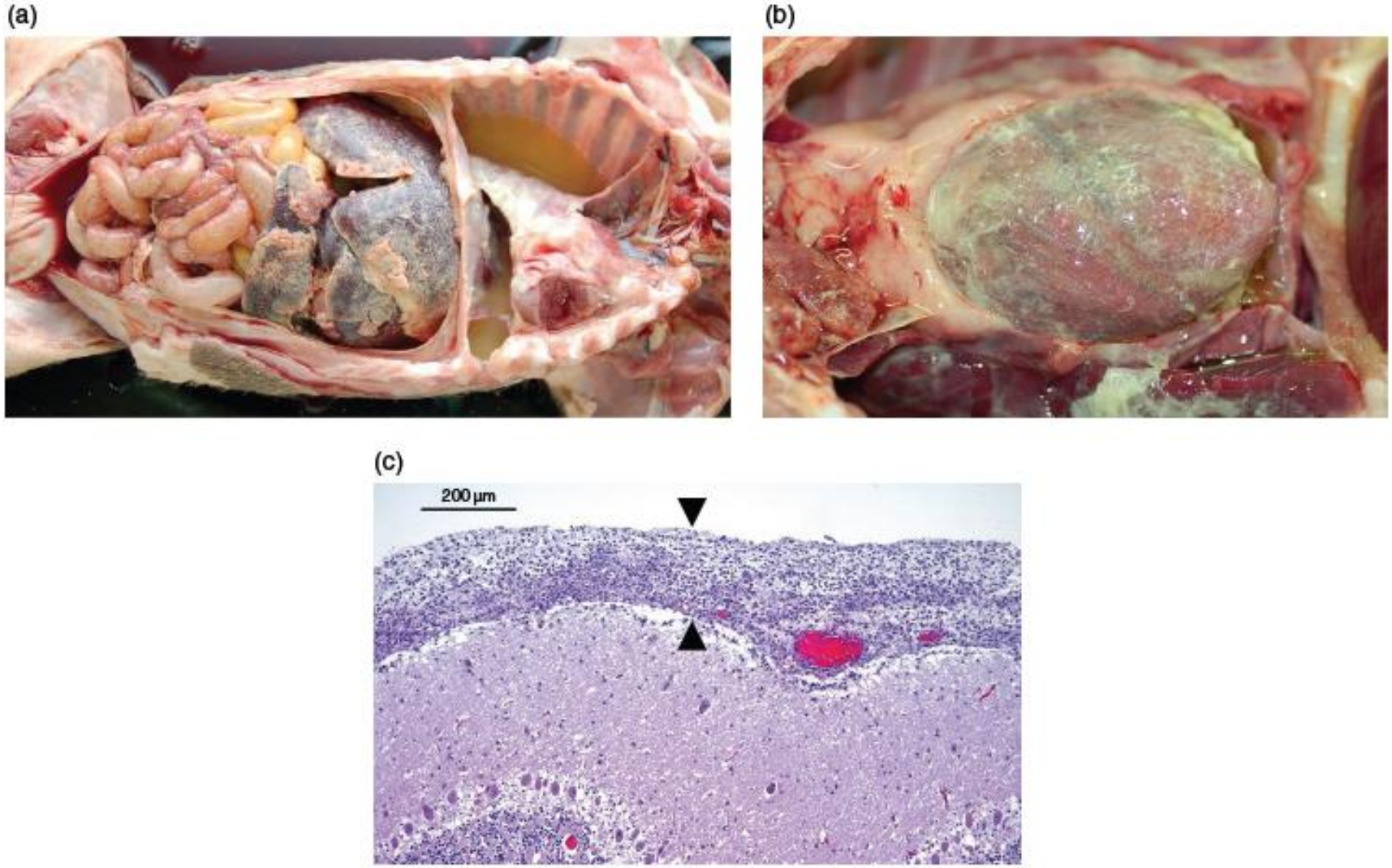
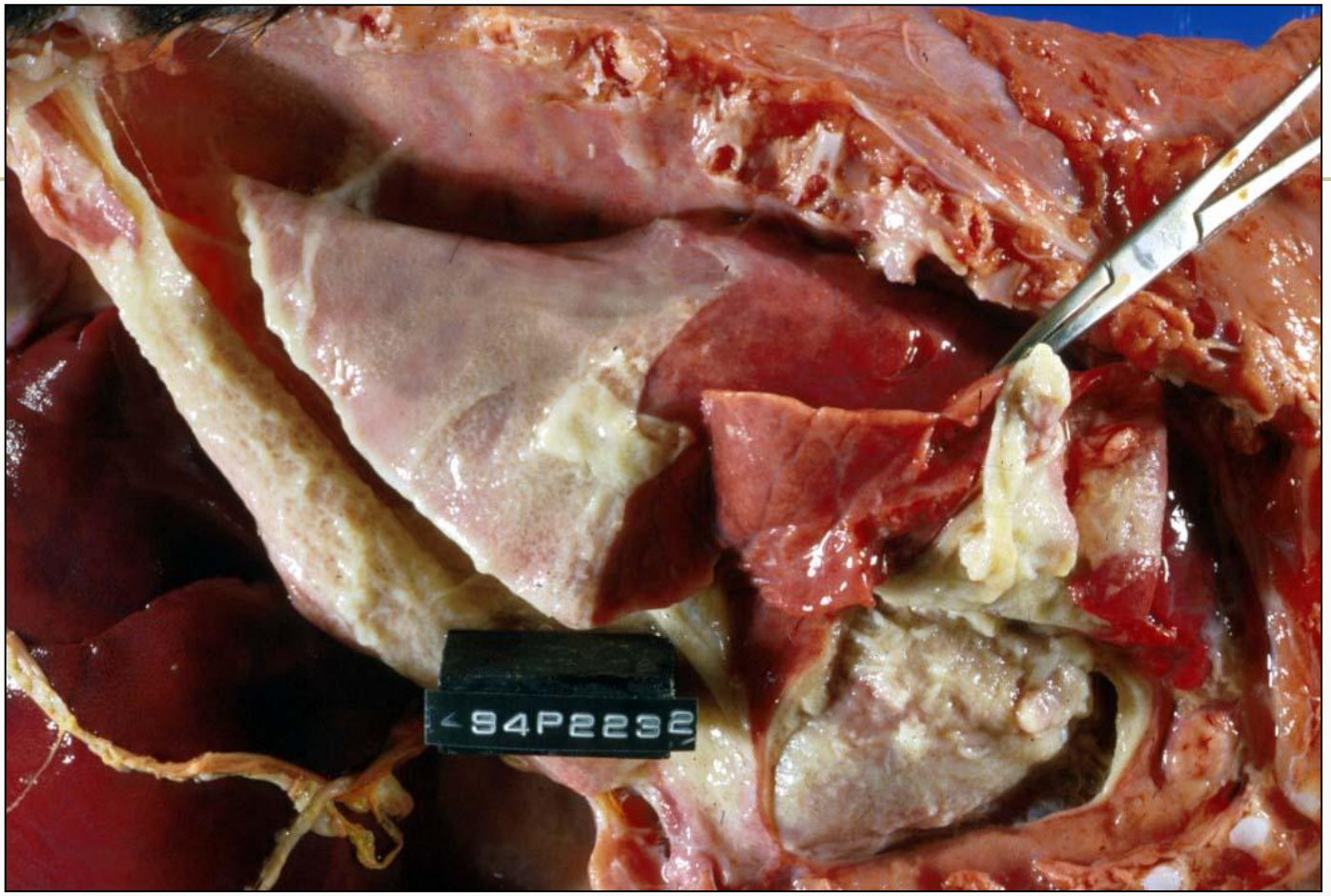


Figure 54.2 Gross and microscopic lesions found in *Haemophilus parasuis* systemic infection. (a) Polyserositis characterized by fibrinopurulent exudate on serosal membranes in peritoneal and thoracic cavities. Note the significant amount of fluid in the thorax. (b) Fibrinopurulent exudate on pericardial surface. (c) Brain, fibrinopurulent meningitis.



纖維素性多發性漿膜炎(fibrinous polyserositis): 胸腔心囊與腹腔的臟器都被黃白色的纖維素性滲出物質粘連

劉振軒等著 豬病與屠宰衛生檢查圖譜 **1997** 台灣養豬科學研究所



Severe, acute, diffuse fibrinopurulent pleuritis and pericarditis due to *H. parasuis*

Courtesy of Dr. Madson, Iowa State University, USA

治療與預防 (Treatment and prevention)

- Early antimicrobial therapy may be effective in clinically affected pigs.
- Antimicrobial susceptibility should be considered because resistance patterns vary among isolates.
- Reduce stress associated with weaning, transport and commingling.
- Control important coinfections.
- Commercial or autogenous vaccines may be useful, but protection may be strain/serovar dependent.

114年度國考試題

下列有關豬格拉瑟病（Glässer's disease）之敘述，何者正確？

A. 致病原為消化道之常在菌叢

B. 常發生在保育期仔豬

C. 常見出血性炎症反應

D. 不感染中樞神經系統

112年度國考試題

下列那一種病原會造成豬隻所謂的Glässer's disease？

A. *Pasteurella multocida*

B. *Haemophilus parasuis*

C. *Salmonella* Typhimurium

D. *Erysipelothrix rhusiopathiae*

111年度國考試題

下列何者為豬隻感染格拉瑟病（Glässer's disease）常見之病變？

A. 腸炎

B. 多發性漿膜炎

C. 肝臟有粟粒狀大的壞死白點

D. 滲出性表皮炎

109年度國家考試題目

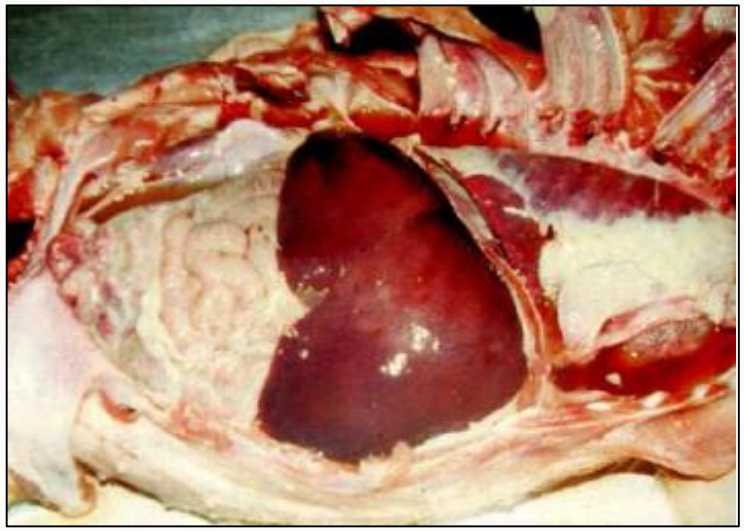
下圖病豬剖檢後可看到胸水及腹水，肺臟、心臟及腸管上均覆蓋多量白色纖維素，最有可能的致病原為下列何者？

A. *Bordetella bronchiseptica*

B. *Klebsiella pneumoniae*

C. *Salmonella* Typhimurium

D. *Haemophilus parasuis*



109年度國考試題

下列何者的培養並不需要X因子，但需要V因子？

A. *Mycoplasma hyopneumoniae*

B. *Erysipelothrix rhusiopathiae*

C. *Salmonella* Choleraesuis

D. *Haemophilus parasuis*

102年度國考試題

下列和者與Glässer's氏病無關?

A. 腹膜炎

B. 關節炎

C. 腦膜炎

D. 肝炎

101年度國考試題

下列有關豬多發性漿膜炎之敘述，何者錯誤?

A. 由嗜血桿菌 (*Haemophilus parasuis*) 感染引起的疾病，又名Glässer's 病

B. 保育豬及肥育豬極少發生

C. 感染豬隻有肺炎及關節炎之病症

D. 感染豬隻有腹膜炎及肺炎之病症

99年度國考試題

下列何者是豬Glässer's disease的病原?

A. *Haemophilus*

B. *Pasteurella*

C. Pseudorabies virus

D. Poxvirus

98年度國考試題

豬的多發性漿膜炎由*Haemophilus parasuis*引起，又稱為什麼病?

A. Greasy病

B. Glässer's氏病

C. Johne氏病

D. Heart water病

96年度國考試題

下列何者又稱Glässer's disease？

A. 水腫病

B. 放線桿菌胸膜肺炎

C. 芽胞梭菌症

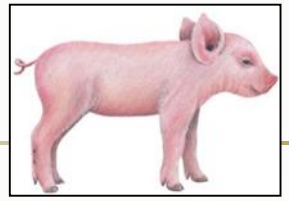
D. 多發性漿膜炎

大腸桿菌感染
(*Escherichia coli* infection)

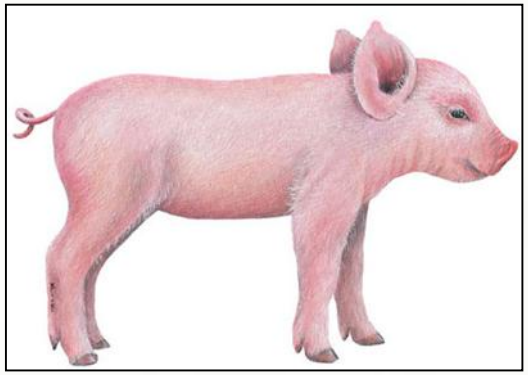
關聯 (Relevance)

All over the world, *E. coli* is an important cause of a wide range of diseases in pigs, including:

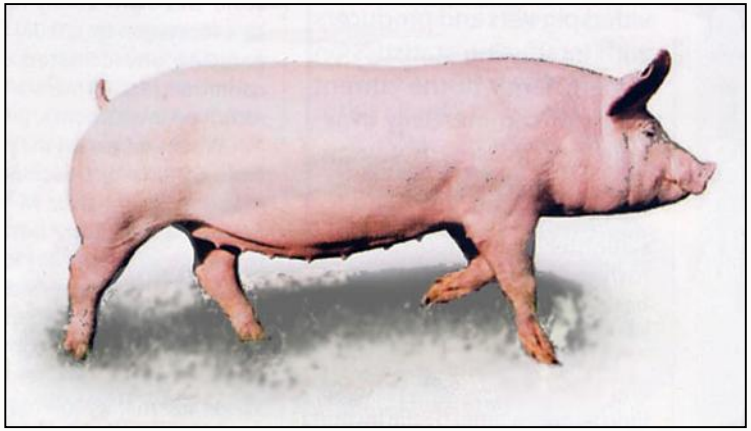
- Neonatal *Escherichia coli* diarrhea (新生仔豬下痢)
- Postweaning *Escherichia coli* diarrhea (離乳後下痢)
- Edema disease (水腫病)
- Mastitis (乳房 炎)
- Urinary tract infection (尿道感 染)



新生仔豬下痢 (neonatal diarrhea)
ETEC



離乳後下痢 (post-weaning diarrhea)
水腫病 (edema disease)
ETEC, AEEC, EDEC



腸桿菌性乳炎 (coliform mastitis)
乳房炎-子宮炎-無乳症候群
(mastitis-metritis-agalactia (MMA) syndrome)

病因 (Etiology)

大腸桿菌 (*Escherichia coli*)

- 屬於腸內細菌科 (*Enterobacteriaceae*)
- 常存於動物腸道，也會隨著腸內容物，散播至環境
- 絕大部分無害，但有一些為病原菌
  - I. enterotoxgenic *E. coli* (ETEC)
  - II. attaching and effacing *E. coli* (AEEC; like enteropathogenic *E. coli* [EPEC] in humans)
  - III. edema disease *E. coli* (EDEC); 也有人稱之Shiga toxin-producing *E. coli* (STEC)

仔豬大腸桿菌下痢 (neonatal *Escherichia coli* infection)

Of all the disease in the sucking piglets, diarrhea is the most common and probably the most important.

Most commonly found in pigs aged from 0-4days

--caused by ETEC

Adherence: Fimbriae F4 (K88), F5 (K99), F6 (987P), or F41

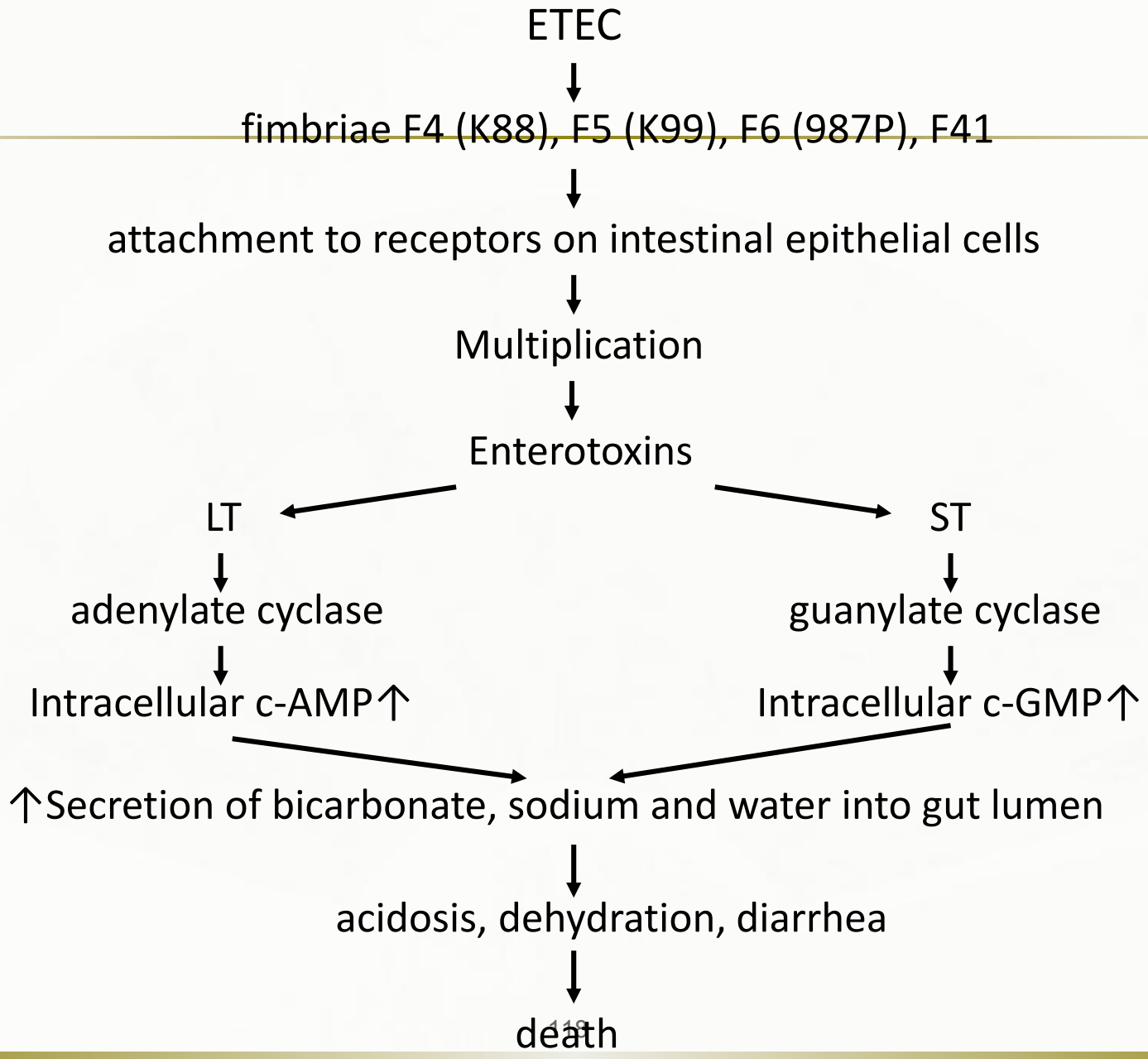
Toxin: Sta, Stb, or LT

LT → adenylate cyclase → ↑cAMP

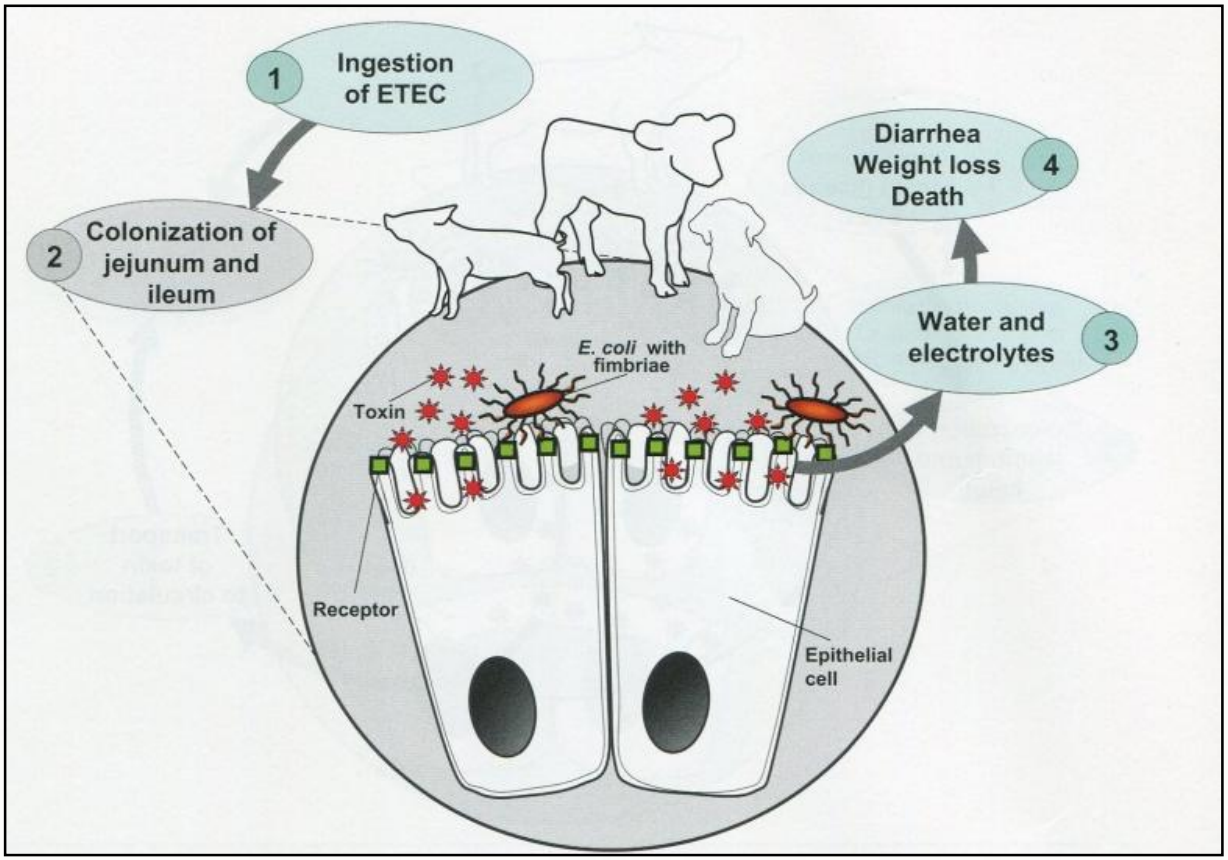
STa → guanylate cyclase → ↑cGMP

→ electrolyte and water secretion→ secretory diarrhea

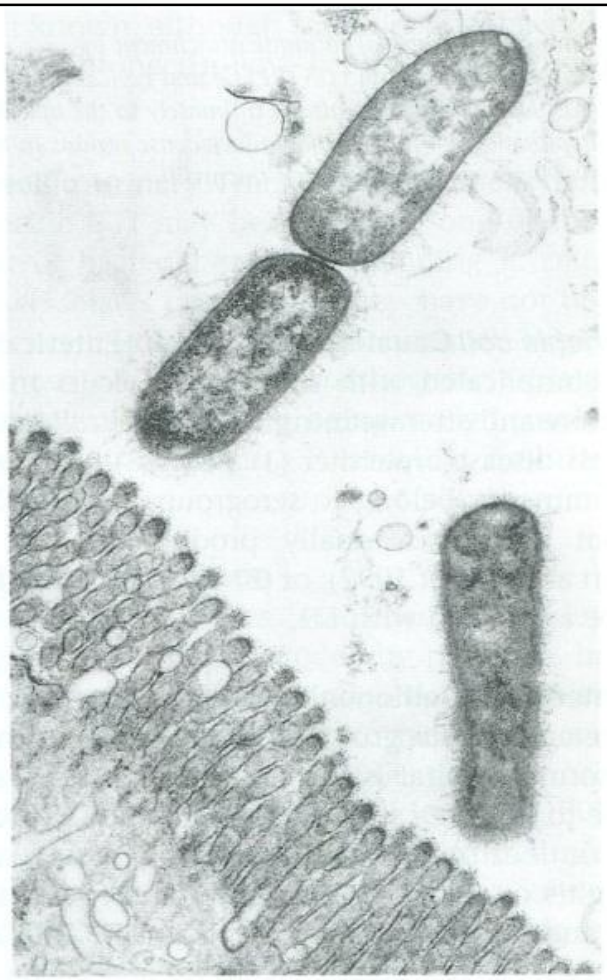
Pathogenesis for ETEC



Schematic representation of the steps involved in the pathogenesis of ETEC infection



Gyles rt al., Pathogenesis of bacterial infections in animals, 2010, Wiley-Blackwell



53.1. *Electron micrograph of fimbrial attachment typical of enterotoxigenic* Escherichia coli *(ETEC) in the intestine. Hairlike fimbriae extend from the surface of bacterial cells that are located approximately half a bacterial width away from the microvilli of intestinal epithelial cells.*







Neonatal piglet showing severe depression caused by *E. coli* enteritis.



Perineum of neonatal piglet suffering from *E. coli* enteritis.

Handbook of Pig Medicine by Jackson and Cockcroft , 2007, Elsevier



脫水、眼窩下陷、削瘦、毛髮粗糙



豬隻畏冷而擠聚成堆



小腸擴張、腸壁變薄且呈透明狀，腸腔內有多量液體蓄積

張聰洲等 經濟動物常見疾病-豬隻篇 2015

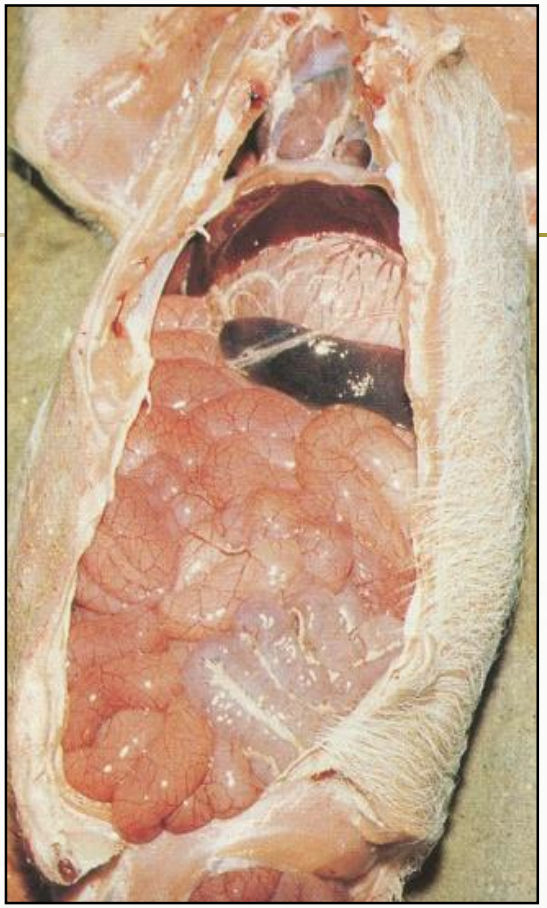


初生仔豬 1-2日齡 下痢脱水致死

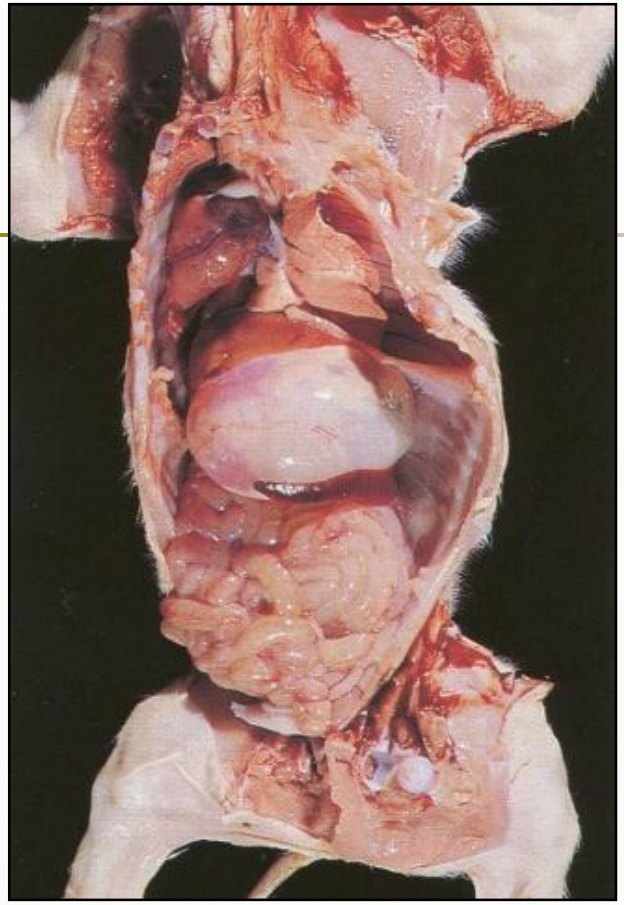


Note the skin is tightly drawn over the ribs and the prominence of the pelvic bones

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

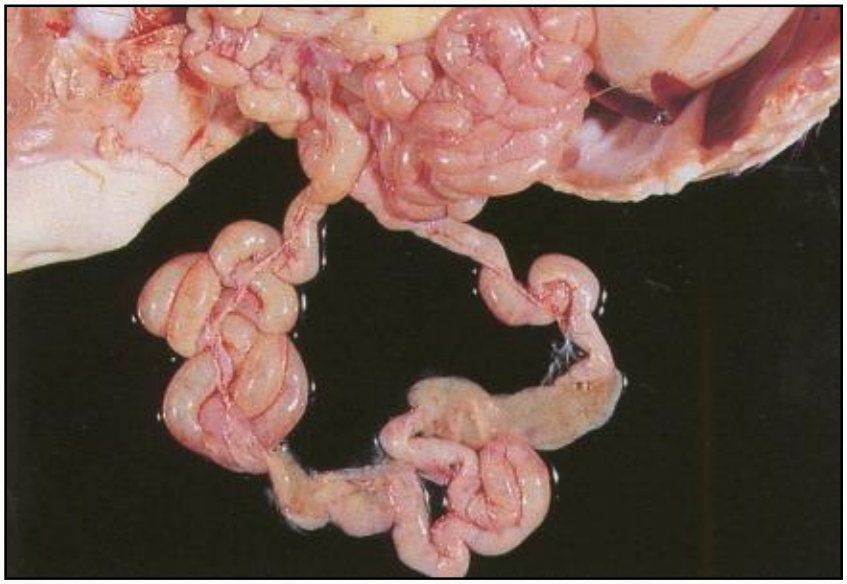


Neonatal *E. coli* infection. Note the pale, fluid-filled intestines. The hair coat is prominent because of the dehydration.



Neonatal *E. coli* infection. The distended stomach filled with milk. The intestines are filled with fluid but not inflamed.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Neonatal *E. coli* infection. Note the pale, uninflamed mucosa and the pale contents .



Neonatal *E. coli* infection. The milk clot in the stomach of effected piglet is frequently found to be laminated.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

離乳後大腸桿菌下痢與水腫病 (postweaning *Escherichia coli* diarrhea and edema disease)

* Postweaning *E. coli* diarrhea (PWD) and edema disease (ED) are often both occur in the same age group of pigs.

* Pathogenesis

  --adherence

  --diarrhea

  --enterotoxemia-Stx2e (相同的名稱如: Shiga-like toxin, verotoxin, edema disease principle, neurotoxin, vasotoxin)

離乳後大腸桿菌下痢與水腫病 (postweaning *Escherichia coli* diarrhea and edema disease)

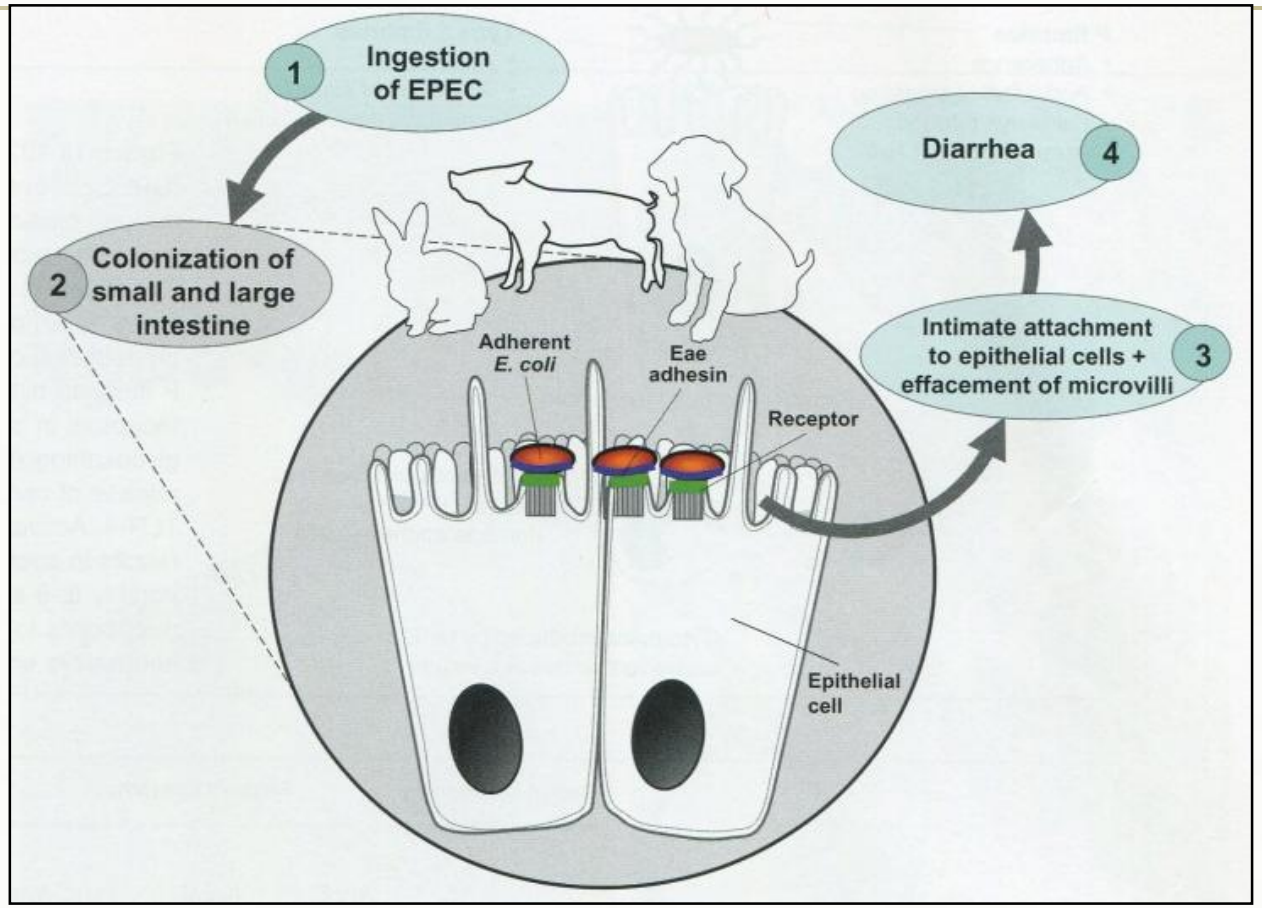
* Older unweaned piglets can be affected by *E. coli* diarrhea and ED as levels of protective antibody wane in maternal milk.

* Loss of milk antibodies at weaning contributes significantly to susceptibility of pigs to *E. coli* enteric infections in the postweaning period. As a result, most disease is observed postweaning.

postweaning *Escherichia coli* diarrhea

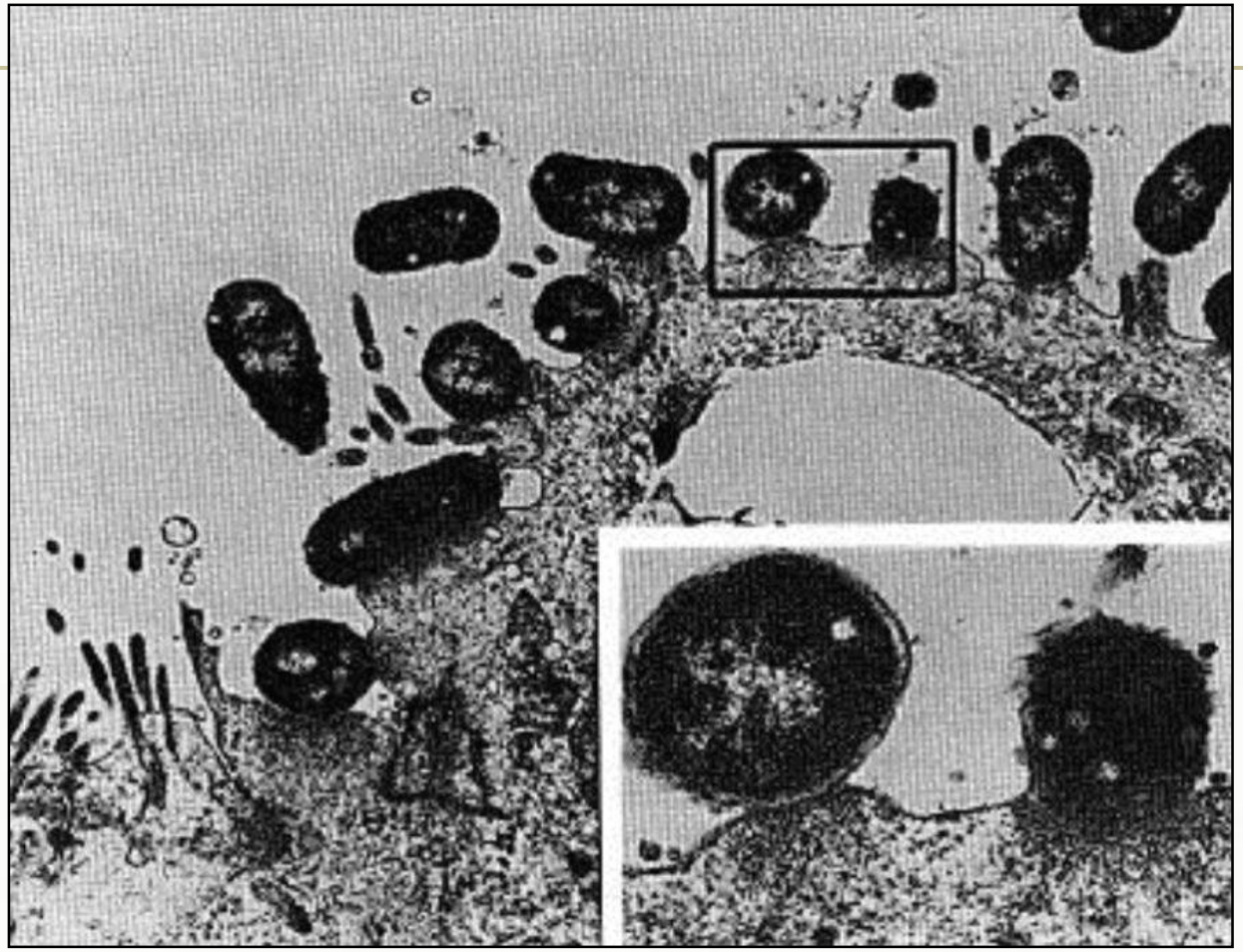
- Postweaning diarrhea (PWD) is mostly commonly caused by ETEC, but can also caused by AEEC (EPEC).
- PWD strains usually have either F4 (K88) or F18 fimbriae.
- AEEC attach intimately to the intestinal epithelial cells. Attached AEEC efface the microvilli and sometimes invade the epithelial cells.

Schematic representation of the steps involved in the pathogenesis of EPEC infection

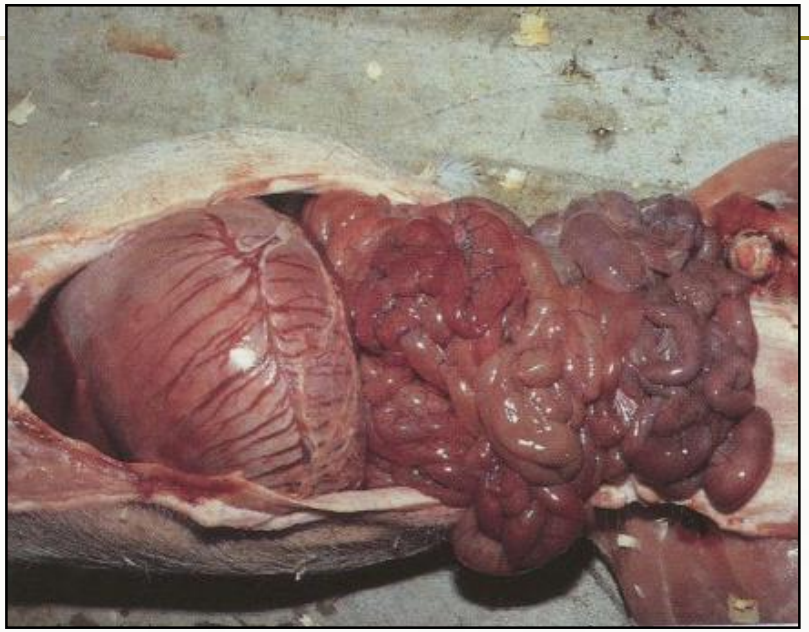


Gyles rt al., Pathogenesis of bacterial infections in animals, 2010, Wiley-Blackwell

EPEC



From: Molecular Genetics of Bacterial Pathogenesis





E. coli post-weaning diarrhea. Stomach filling is sometimes seen, and the congestion of the small intestine is often noted.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Post-mortem appearance of a growing pig suffering from acute postweaning enteritis. Note severe inflammation of the fluid-filled small intestine.

Handbook of Pig Medicine by Jackson and Cockcroft , 2007, Elsevier

Postweaning *Escherichia coli* diarrhea

- Prevention and control

Weaning management

Hygiene and environmental control

Diet

Vaccination/immunoprophylaxis

Genetic resistance

Targeted antimicrobial therapy when necessary
Fluid/electrolyte support

Edema disease
(other names as gut edema or bowel edema)

- Edema disease is caused by hemolytic *E. coli* that produce F18 fimbriae and Shiga toxin 2e.

- Peracute toxemia caused by *E. coli* that affects primarily healthy, rapidly growing nursery pigs.

- Shiga toxin 2e produced in the intestine of colonized pigs is responsible for the major clinical signs and pathology that are observed.

Edema disease
(other names as gut edema or bowel edema)

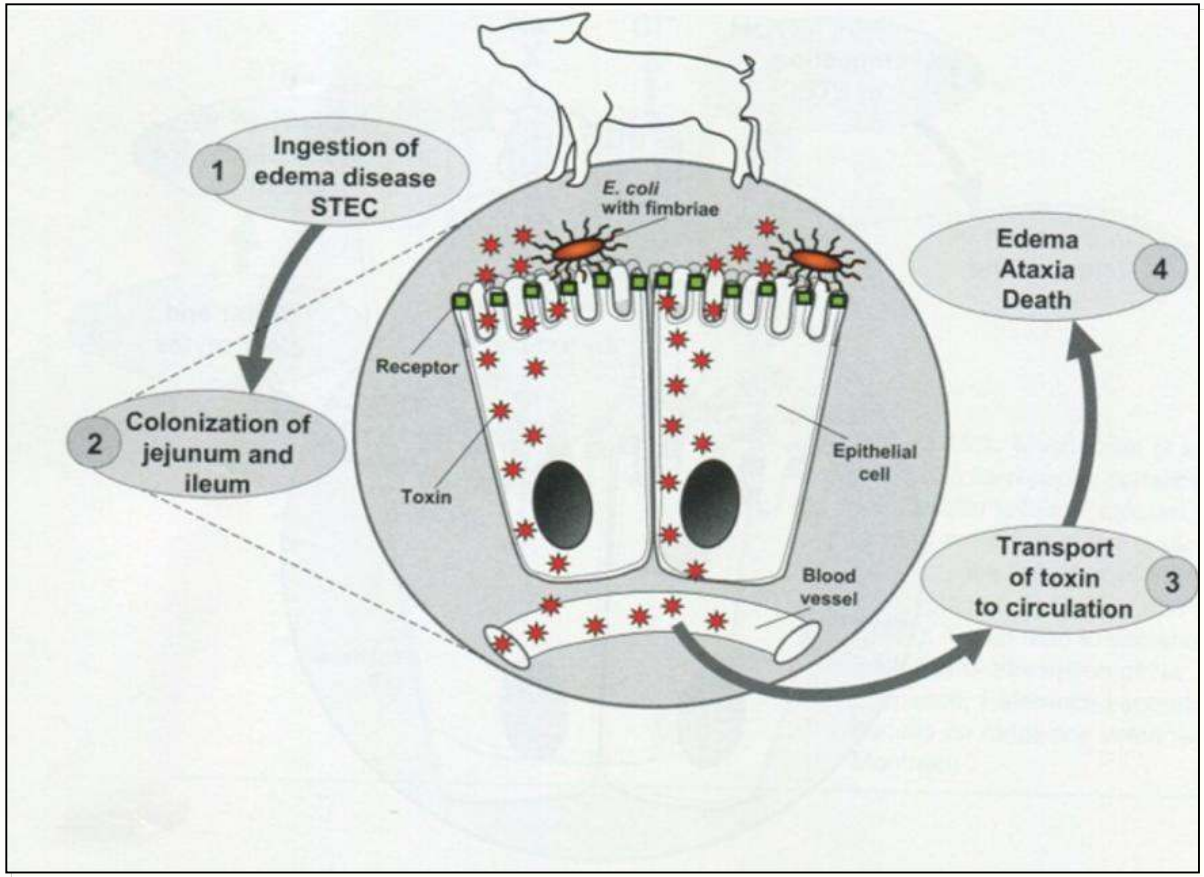
- The toxin is absorbed from the intestine and targets vascular endothelium in specific sites believed to have high concentrations of the toxin receptor globotetraosyl ceramide.

- The abovementioned sites include the submucosa of the stomach, the colonic mesentery, the subcutaneous tissues of the forehead and eyelids, the larynx, and the brain.

- Damage to vascular endothelium results in edema, hemorrhage, intravascular coagulation, and microthrombosis.

- Prominent feature of ED: edema of the submucosa of the stomach and the mescolon (腸繫膜).

Schematic representation of the steps involved in the pathogenesis of STEC infection in edema disease of pigs



Gyles rt al., Pathogenesis of bacterial infections in animals, 2010, Wiley-Blackwell

Edema disease
(other names as gut edema or bowel edema)

* Control of edema disease

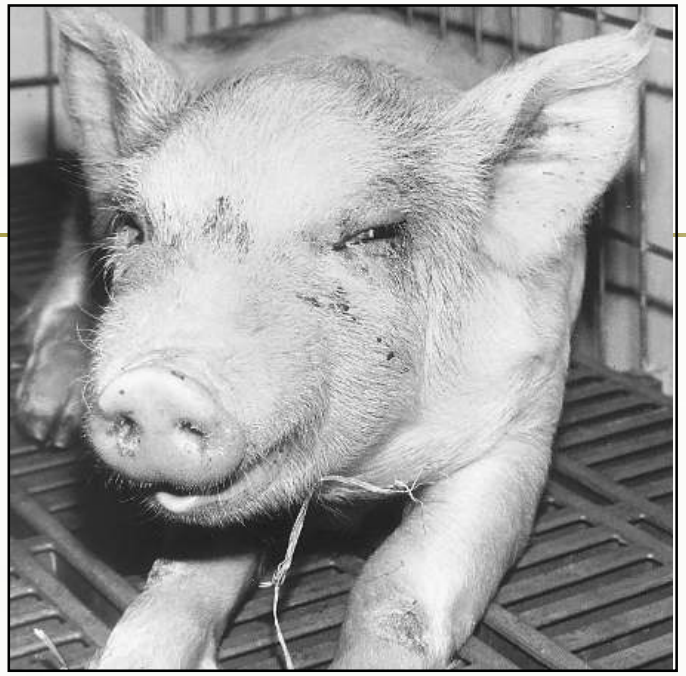
Prevention is more important than treatment because clinical disease may progress rapidly.

Reduce weaning stress and abrupt dietary changes.

Vaccination against Stx2e/F18-associated disease may be used where appropriate.

Antimicrobial use should be targeted and based on clinical and susceptibility information whenever possible.



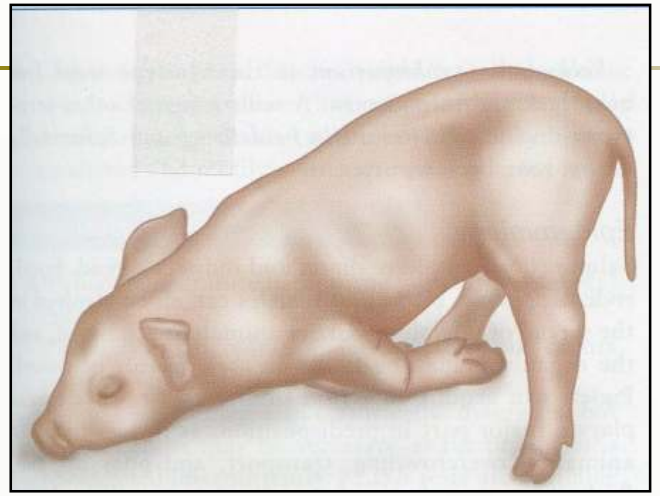


Edematous swelling of eyelids, forehead, and lips





Note the puffy eyelids due to edema and the subcutaneous edema revealed by the incision

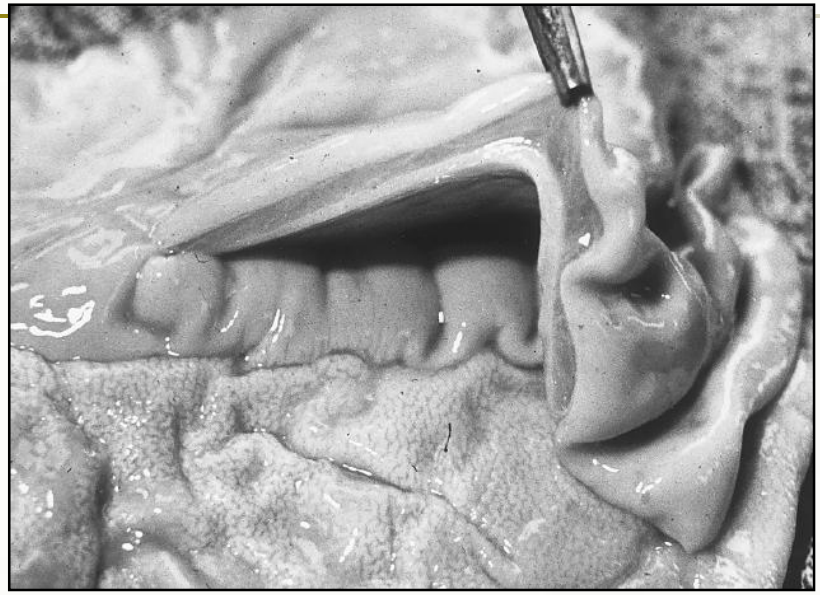


Note the flexion of the forelimbs and the eye closed by an edematous upper eyelid

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



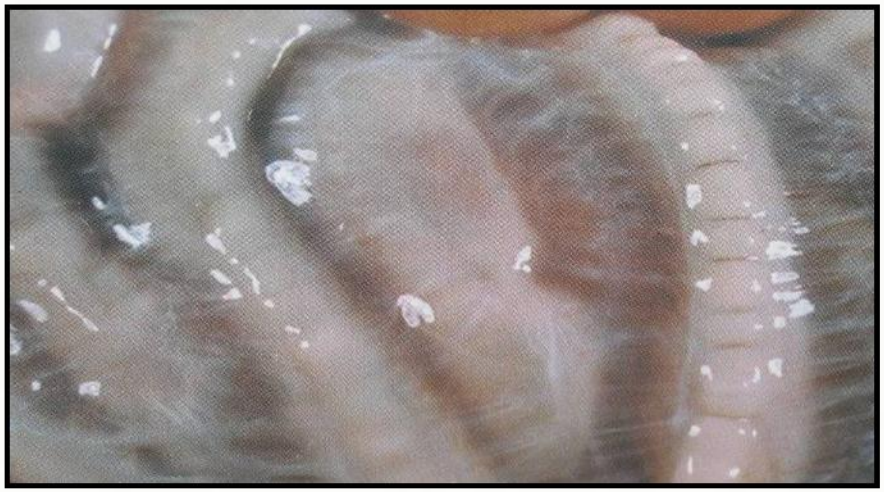
Edema of mescolon



Edema of submucosa of the stomach

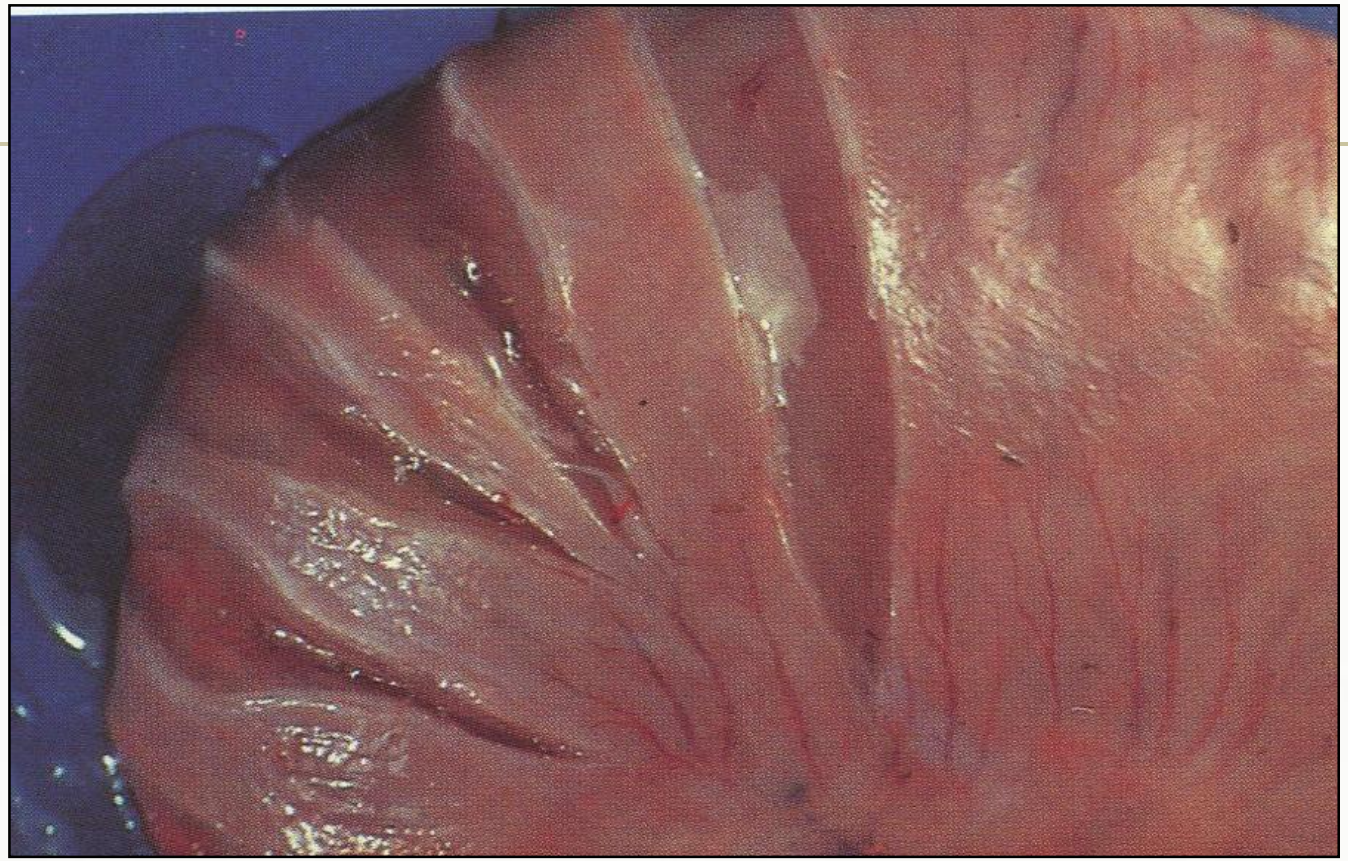


Gastric wall edema



Mesocolonic edema

Courtesy of Dr. Madson, Iowa State University, USA



The wall of this stomach has been cut to demonstrate the edema. Fluid can be seen leaking from the incisions.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

Mastitis

- The intestinal flora of the sow, the oral flora of the neonatal piglet, and environmental bacteria may significantly contribute to contamination of nipples.

- The bedding of the sow is very important, permitting dung and urine to contaminate the udder.

- Clinical signs include pyrexia, listlessness, weakness, lost of interest in the piglets.

Mastitis

* The behavior of the piglets is very helpful in the early detection of lactational failure. Undernourished piglets look gaunt. They frequently try to suck, move from nipple to nipple, nibble at litter, and lick urine from the floor.

* Lesions of mastitis are confined to the mammary glands and regional lymph nodes.



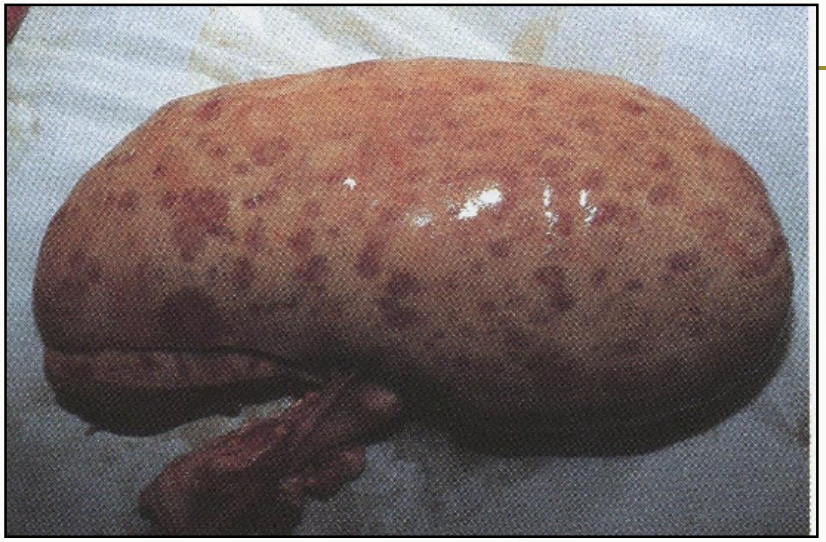


E. coli mastitis

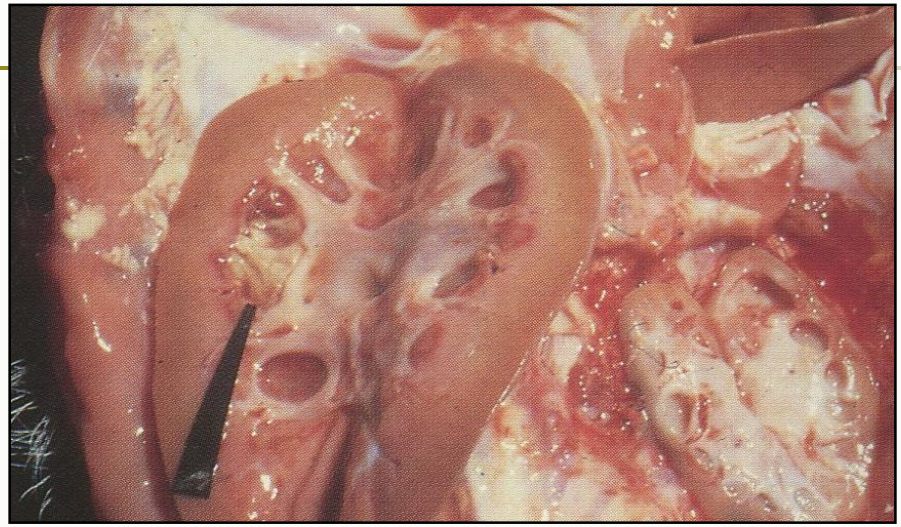
Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

Urinary Tract Infection (UTI)

- Noncontagious infection of endogenous origin.
- Contamination and ascent of the urethra by fecal flora is more likely in females than in males.
- Under intensive confinement conditions, sow's vulvas are often placed in direct contact with feces; the dog-sitting position helps to force fecal material into the vagina.
- *E. coli* producing type 1 fimbriae and P fimbriae facilitates UTI.



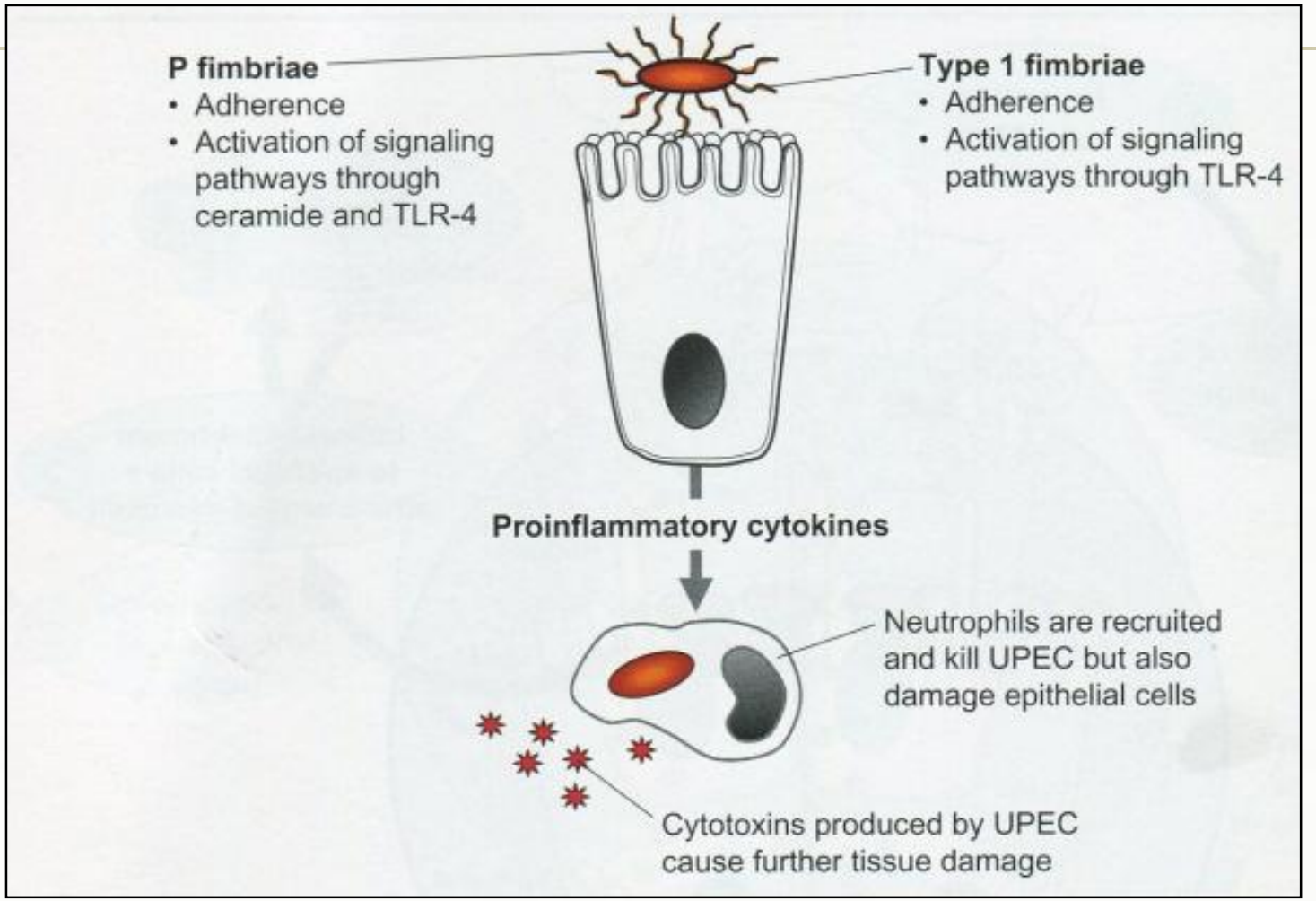
E. coli pyelonephritis. Note the mottled surface of the kidney.



A kidney of a four-year-old sow opened to demonstrate the changes in pyelonephritis. Note the pus in the renal pelvic (black strip)

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

Uropathogenic *E. coli* (UPEC) in cystitis



Gyles rt al., Pathogenesis of bacterial infections in animals, 2010, Wiley-Blackwell

115年度國家考試題目

大腸桿菌所引起的豬原發性敗血症，好發於下列何種階段？

A. 1～4 日齡

B. 3～4 週齡

C. 5～6 週齡

D. 7～8 週齡

115年度國家考試題目

病原性大腸桿菌所產生的heat-labile toxin，其作用機制最接近下列那一個毒素？

A. Shiga toxin

B. cholera toxin

C. listeriolysin

D. suilysin

115年度國家考試題目

下列何種病原可造成豬隻過度分泌型（hypersecretion）下痢？

A. *Brachyspira hyodysenteriae*

B. *Lawsonia intracellularis*

C. *Escherichia coli*

D. *Clostridium perfringens* Type C

114年度國家考試題目

下列何種豬致病型大腸桿菌，會造成豬隻腸上皮細胞attaching and effacing病灶?

A. Enterotoxigenic *E. coli*（ETEC）

B. Enteropathogenic *E. coli*（EPEC）

C. Extraintestinal pathogenic *E. coli*（ExPEC）

D. Enteroaggregative *E. coli*（EAggEC）？

113年度國考試題

利用鑑定*Escherichia coli*表面不同的O、H、K或F抗原，可用來進行血清分型，其中H代表那一種抗原？

A. 線毛抗原

B. 莢膜抗原

C. 菌體抗原

D. 鞭毛抗原

112年度國考試題

下列何種病原性大腸桿菌可造成剛離乳保育豬之水腫病？

A. enteroaggregative *Escherichia coli*

B. enterotoxigenic *Escherichia coli*

C. enteropathogenic *Escherichia coli*

D. Shiga toxin-producing *Escherichia coli*

112年度國考試題

大腸桿菌除線毛外，下列何種因子是造成豬水腫病的主要原因？

A. 莢膜

B. 毒素

C. 脂多醣

D. 鞭毛

110年度國考試題

造成新生仔豬下痢的大腸桿菌通常是帶有K88、K99或987P抗原的enterotoxigenic *Escherichia coli*。上述的抗原是細菌的何種構造或成分？

A.莢膜（capsule）

B.鞭毛（flagella）

C.孔蛋白（porin）

D.線毛（fimbriae）

110年度國考試題

下圖為離乳後1～3週的豬隻，眼瞼水腫且剖檢後在胃壁的黏膜下層也看到膠樣水腫。可能造成此症狀及病變的病原為何？

A. *Staphylococcus hyicus*

B. *Escherichia coli*

C. *Streptococcus suis*

D. *Haemophilus parasuis*



107年度國考試題

下列有關*Escherichia coli*造成的新生仔豬下痢，何者正確？

A.由enteropathogenic *Escherichia coli* strain所造成

B.腸黏膜破壞嚴重，所以下痢便通常帶有血液及黏液

C.病原通常帶有線毛及腸毒素二個重要致病因子

D.通常發生在1週齡以上的仔豬

107年度國考試題

下列何者為0—4日齡新生仔豬下痢最常見的病原？

A. *Salmonella* Typhimurium

B. *Isospora suis*

C. *Brachyspira hyodysenteriae*

D. *Escherichia coli*

106年度國考試題

下列有關豬隻水腫病（edema disease）的敘述，何者正確？

A.常發生於一週齡以下仔豬

B.豬隻一旦發病，通常預後不佳，死亡率高

C.病原為*Brucella suis*

D.致病機制主要是因為病原分泌一種造成細胞cyclic AMP上升的毒素所導致

105年度國考試題

腸毒素產生型大腸桿菌（enterotoxgenic *Escherichia coli*，ETEC）會造成新生仔豬下痢（neonatal diarrhea），最重要的致病因子是產生毒素以及下列何種抗原？

A.線毛抗原（fimbrial antigen）

B.莢膜抗原（capsular antigen）

C.鞭毛抗原（flagellar antigen）

D.體抗原（somatic antigen）

104年度國考試題

下列有關豬隻大腸桿菌症之敘述，何者錯誤？

A. 造成初生仔豬下痢

B. 造成離乳豬下痢

C. 造成保育豬水腫病

D. 造成肥育豬血痢

104年度國考試題

引起豬尿道感染的大腸桿菌最常看到帶有P線毛(P fimbriae）及下列那一種線毛？

A. F4

B. F5

C. curli

D. type 1

103年度國考試題

下列有關豬隻水腫病（edema disease）的敘述，何者正確？

A. 常發生於離乳前仔豬

B. 水腫較常見於胃黏膜下層及腸繫膜

C. 病原為*Clostridium perfringens*

D. 病原在blood agar上通常不具溶血性

101年度國考試題

有關水腫病（edema disease）的敘述，下列何者正確？

A.主要發生在豬，致病因子為大腸桿菌毒素

B.病原為梭狀桿菌，主要發生在羊

C.結腸與胃壁的水腫最不明顯

D.為台灣常見的疾病

100年度國考試題

有關豬大腸桿菌症的敘述，下列何項錯誤？

A.易發生在出生至離乳後2—3週的仔豬

B.腸管絨毛嚴重萎縮

C.疾病發生無季節性

D.剖檢胃內常見凝乳塊，小腸失去彈性

99年度國考試題

下列有關豬水腫病的敘述，何者錯誤？

A.罹病後因毒素侵害血管內皮細胞而水腫

B.常發生於離乳後到保育期

C.罹病後有急性發熱現象

D.與溶血性大腸桿菌有關